Softwarica
College of IT & E-commerce

Town Recommendation system

Data Science for Developer- ST5014CEM

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Abstract

This report aims to provide a data-driven recommendation to assist international investors in selecting the ideal region between Devon and Dorset for their property purchase in the UK. While both regions boast stunning landscapes, rich cultural heritage, and unique charm, the decision-making process extends beyond aesthetics. Key factors such as affordability, transportation connectivity, safety, and overall quality of life have been meticulously analyzed to ensure a comprehensive evaluation. By leveraging quantitative and qualitative insights, this analysis seeks to highlight the strengths and weaknesses of each region, empowering the investors to make an informed and confident choice aligned with their long-term goals.

Keywords

Figure 1 : Keywords

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Introduction

Investing in property is a significant decision, especially for international buyers seeking a balance between affordability, connectivity, safety, and quality of life. This Town Recommendation System evaluates towns in Devon and Dorset using key metrics like housing prices, broadband speed, and crime rates, along with an additional characteristic, to provide a data-driven comparison. Developed in R and based on UK government datasets, the system normalizes data to 3NF for accuracy and generates a recommendation score (0–10) for each town. By highlighting the top three towns, this system empowers informed property investment decisions in these picturesque UK regions.

Cleaning Data

The data preprocessing phase was essential to ensure the datasets were accurate, consistent, and suitable for analysis. Missing values were handled using imputation techniques such as mean, median, or mode substitution, or were removed if they significantly compromised data quality. Duplicate records were identified using primary keys or composite keys and eliminated to maintain data integrity. Outliers were detected using statistical methods such as the Z-score, IQR, or visualization techniques and were either transformed, capped, or removed to prevent skewed results.

The data was normalized to the third normal form (3NF) to eliminate redundancy and ensure referential integrity across relational tables. Categorical variables were encoded using techniques like one-hot encoding or label encoding, ensuring compatibility with downstream algorithms. Date fields were parsed and standardized into appropriate datetime formats, enabling accurate time-series analysis. Numerical attributes underwent rigorous validation, including range checks, scaling (e.g., normalization or standardization), and type conversion, to ensure consistency and eliminate anomalies. This systematic and technical approach established a robust, error-free dataset for reliable analytical and modeling tasks.

House sales

Obtained Data

	{CB0035E6-3546-58AE-E053-6B04A8C091AF}	630000	2021-04-29 00:00	BN13 3AH	D	N	F	3	...9	HIGHLANDS CLOSE	...11
1	{CB0035E6-3547-58AE-E053-6B04A8C091AF}	477000	2021-05-10	BN16 2PQ	D	N	F	10	NA	CHAUCER AVENUE	RUSTINGTO
2	{CB0035E6-3548-58AE-E053-6B04A8C091AF}	370000	2021-05-06	BN2 4HZ	T	N	F	6	NA	DARTMOUTH CLOSE	NA
3	{CB0035E6-3549-58AE-E053-6B04A8C091AF}	462500	2021-04-28	BN2 0GP	T	N	F	18	NA	STANLEY STREET	NA
4	{CB0035E6-354A-58AE-E053-6B04A8C091AF}	433000	2021-08-06	BN8 4LS	D	N	F	11	NA	POWELL ROAD	NEWICK
5	{CB0035E6-354B-58AE-E053-6B04A8C091AF}	676000	2021-08-05	BN1 6NB	T	N	F	35	NA	BALFOUR ROAD	NA
6	{CB0035E6-354D-58AE-E053-6B04A8C091AF}	367000	2021-02-24	RH10 6RY	S	N	F	19	NA	GREENACRES	FURNACE G
7	{CB0035E6-354E-58AE-E053-6B04A8C091AF}	418000	2021-04-08	BN8 4LS	D	N	F	18	NA	POWELL ROAD	NEWICK
8	{CB0035E6-354F-58AE-E053-6B04A8C091AF}	425000	2021-05-07	BN13 3LG	D	N	F	23	NA	CLYDE ROAD	NA
9	{CB0035E6-3550-58AE-E053-6B04A8C091AF}	390000	2021-06-22	BN2 9XS	T	N	F	31	NA	ARNOLD STREET	NA
10	{CB0035E6-3551-58AE-E053-6B04A8C091AF}	420000	2021-06-25	BN16 2JR	T	N	F	34	NA	MALLON DENE	RUSTINGTO
11	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Figure 2: Obtained House Dataset

Cleaning Code

```r
#load the Libraries

library(tidyverse)
library(dplyr)

#For Date manupulation

library(lubridate)

# Using tidyverse for reading csv file and assigning to variable
house_data_2021<- read_csv("ObtainedData/HousePrice21.csv")
house_data_2022 <- read_csv("ObtainedData/HousePrice22.csv")
house_data_2023 <- read_csv("ObtainedData/HousePrice23.csv")

# Adding columns for all first data frame
names(house_data_2021)=columns <- c(
  "UniqueId",
  "Price",
  "DateOfTransaction",
  "Postcode",
  "PropertyType",
  "NewBuild",
  "Leasehold",
  "StreetName",
  "FlatNumber",
  "StreetAddress",
  "Locality",
  "Town",
  "District",
  "County",
  "AdditionalPropertyInformation",
  "AdditionalColumn"
)

# Assigning column names of house_data_2021 to house_data_2022 and house_data_2023

colnames(house_data_2022)=colnames(house_data_2021)
colnames(house_data_2023)=colnames(house_data_2021)
View(house_data_2022)
View(house_data_2023)
```

```r
# merging the data frame to make one single dataframe and rechecking the no of rows
houseprice=rbind(house_data_2021,house_data_2022,house_data_2023)
View(houseprice)
dim(houseprice)



houseprice=houseprice %>%
  mutate(DateOfTransaction = ymd(DateOfTransaction)) %>%
  select(DateOfTransaction,Price,County,Town,Postcode)

str(houseprice)

#Writing merged data frame of house price in cleaned folder
write_csv(houseprice,"CleanedData/House/HousePrice(2021-2023).csv")

#Filtering data of dorset of only and  writing new csv
Dorset=houseprice %>%
  filter(County=="DORSET") %>%
  select(DateOfTransaction,Price,County,Town,Postcode)
str(Dorset)
write_csv(Dorset,"CleanedData/House/DorsetHousePrice.csv")


# Filtering House Price for Devon only and writing new cleaned csv
Devon=houseprice %>%
  filter(County=="DEVON") %>%
  select(DateOfTransaction,Price,County,Town,Postcode)
write_csv(Devon,"CleanedData/House/DevonHousePrice.csv")
View(Devon)

# Making table with unique postal code and town as reference to broadband data frame to figure out Devon and Dorset towns from that table

postcode_town <- houseprice %>%
  filter((County == "DEVON" | County == "DORSET") & !is.na(Postcode)) %>%
  select(Postcode, Town, County) %>%
  distinct() %>%
  group_by(Postcode)

write_csv(postcode_town,"CleanedData/Postal/PostalCodeTable.csv")
```

(Top Level) ÷ R Scrip

Figure 3: House Sales cleaning

Cleaned Data

	UniqueId	Price	DateOfTransaction	Postcode	PropertyType	NewBuild	Leasehold	StreetName	FlatNumber	StreetAddress
1	{CB0035E6-3547-58AE-E053-6B04A8C091AF}	477000	2021-05-10	BN16 2PQ	D	N	F	10	NA	CHAUCER AVENUE
2	{CB0035E6-3548-58AE-E053-6B04A8C091AF}	370000	2021-05-06	BN2 4HZ	T	N	F	6	NA	DARTMOUTH CLO
3	{CB0035E6-3549-58AE-E053-6B04A8C091AF}	462500	2021-04-28	BN2 0GP	T	N	F	18	NA	STANLEY STREET
4	{CB0035E6-354A-58AE-E053-6B04A8C091AF}	433000	2021-08-06	BN8 4LS	D	N	F	11	NA	POWELL ROAD
5	{CB0035E6-354B-58AE-E053-6B04A8C091AF}	676000	2021-08-05	BN1 6NB	T	N	F	35	NA	BALFOUR ROAD
6	{CB0035E6-354D-58AE-E053-6B04A8C091AF}	367000	2021-02-24	RH10 6RY	S	N	F	19	NA	GREENACRES
7	{CB0035E6-354E-58AE-E053-6B04A8C091AF}	418000	2021-04-08	BN8 4LS	D	N	F	18	NA	POWELL ROAD
8	{CB0035E6-354F-58AE-E053-6B04A8C091AF}	425000	2021-05-07	BN13 3LG	D	N	F	23	NA	CLYDE ROAD
9	{CB0035E6-3550-58AE-E053-6B04A8C091AF}	390000	2021-06-22	BN2 9XS	T	N	F	31	NA	ARNOLD STREET
10	{CB0035E6-3551-58AE-E053-6B04A8C091AF}	420000	2021-06-25	BN16 2JR	T	N	F	34	NA	MALLON DENE
11	{CB0035E6-3552-58AE-E053-6B04A8C091AF}	750000	2021-05-06	BN3 5SA	D	N	F	23	NA	PORTLAND VILLAS
12	{CB0035E6-3553-58AE-E053-6B04A8C091AF}	595000	2021-06-04	BN6 9XZ	S	N	F	11	NA	IDEN HURST
13	{CB0035E6-3554-58AE-E053-6B04A8C091AF}	307000	2021-05-14	BN41 1PA	T	N	F	6	NA	ST RICHARDS ROA
14	{CB0035E6-3555-58AE-E053-6B04A8C091AF}	275000	2021-06-30	PO21 4PQ	T	N	F	130	NA	THE CAUSEWAY
15	{CB0035E6-3556-58AE-E053-6B04A8C091AF}	2900000	2021-05-24	BN3 2JE	T	N	F	8	NA	ADELAIDE CRESCE
16	{CB0035E6-3557-58AE-E053-6B04A8C091AF}	230000	2021-04-19	BN3 3RG	F	N	L	43	GROUND FLOOR FLAT	GOLDSTONE ROAI
17	{CB0035E6-3558-58AE-E053-6B04A8C091AF}	227000	2021-05-13	BN27 1TX	S	N	F	15	NA	FERN GREEN
18	{CB0035E6-3559-58AE-E053-6B04A8C091AF}	1500000	2021-04-30	BN3 6HG	T	N	F	34	NA	HOVE PARK VILLAS
19	{CB0035E6-355B-58AE-E053-6B04A8C091AF}	465000	2021-05-07	BN2 6DJ	S	N	F	48	NA	DOWNLAND ROAI
20	{CB0035E6-355C-58AE-E053-6B04A8C091AF}	555000	2021-08-06	BN5 9JA	D	N	F	49	NA	FURNERS MEAD

Figure 4 : House clean data

Description

The dataset for UK house prices was cleaned to ensure accuracy and consistency. The house price data for 2021, 2022, and 2023 was loaded using read_csv() and combined into a single, cohesive dataset. Column names were standardized and renamed with descriptive labels for clarity. The dataset was meticulously cleaned by removing duplicate and unnecessary columns, trimming whitespace, and ensuring numeric formatting for fields like "Price" and "HouseNumber." To focus on specific regions, records were filtered to include only properties located in Devon and Dorset. Monthly average house prices were calculated by grouping the data by month and averaging sale prices, and these averages were integrated into the main dataset. Exclusive analysis was conducted for towns within Devon and Dorset by grouping data by town and calculating average prices. Separate datasets for Devon and Dorset towns were saved for deeper insights, and additional filtering was performed for 2022 sales in Devon to refine the analysis further. This streamlined process resulted in a well-structured, reliable dataset that supports accurate regional comparisons and insights.

Broadband Speed

Obtained Data

	postcode	postcode_space	postcode area	Median download speed (Mbit/s)	Average download speed (Mbit/s)	Minimum download speed (Mbit/s)	Maximum download speed (Mbit/s)
1	AB101AU	AB10 1AU	AB	15.6	17.4	8.1	72.2
2	AB101BA	AB10 1BA	AB	19.7	18.8	7.8	23.8
3	AB101BB	AB10 1BB	AB	13.9	17.3	4.7	69.1
4	AB101BD	AB10 1BD	AB	13.3	12.9	8.1	16.2
5	AB101FG	AB10 1FG	AB	17.7	19.8	14.3	69.0
6	AB101FL	AB10 1FL	AB	18.1	17.3	11.5	22.0
7	AB101HF	AB10 1HF	AB	14.3	13.4	3.7	18.2
8	AB101HP	AB10 1HP	AB	16.4	15.9	12.9	19.6
9	AB101HW	AB10 1HW	AB	14.6	18.3	3.9	80.0
10	AB101JE	AB10 1JE	AB	13.6	19.2	4.6	69.8
11	AB101JG	AB10 1JG	AB	14.3	16.4	4.9	55.0
12	AB101JJ	AB10 1JJ	AB	12.2	15.0	5.1	51.7

Figure 5 : Broadband obtain data

```r
#Library for using necessary functions
library(tidyverse)
library(dplyr)
library(lubridate)

#Reading all the dataset for broadband and Postalcode

broadband <- read_csv("ObtainedData/broadband1.csv")
postal <- read_csv("CleanedData/Postal/PostalCodeTable.csv")

#Performing left join to bring corresponding column town and county to broadband data frame
merged_data = broadband %>%
  left_join(postal, by = c("postcode_space" = "Postcode"))

# Filtering data that is needed which is of county Devon and Dorset
devonNdorset= merged_data %>%
  filter(County=="DEVON"|County=="DORSET")

# Selecting only necessary column for Visualization and making new data frame
broadband=devonNdorset %>%
  select(postcode_space,`Average download speed (Mbit/s)`,`Minimum download speed (Mbit/s)`,`Maximum download speed (Mbit/s)`,County,Town)


# Writing new csv file for visualization
write_csv(broadband,"CleanedData/Broadband/CleanedBroadband.csv")
```

Figure 6 : Broadband Cleaning Code

Cleaning Code

```
#Library for using necessary functions
library(tidyverse)
library(dplyr)
library(lubridate)

#Reading all the dataset for broadband and Postalcode

broadband <- read_csv("ObtainedData/broadband1.csv")
postal <- read_csv("CleanedData/Postal/PostalCodeTable.csv")

#Performing left join to bring corresponding column town and county to broadband data frame
merged_data = broadband %>%
  left_join(postal, by = c("postcode_space" = "Postcode"))

# Filtering data that is needed which is of county Devon and Dorset
devonNdorset= merged_data %>%
  filter(County=="DEVON"|County=="DORSET")

# Selecting only necessary column for Visualization and making new data frame
broadband=devonNdorset %>%
  select(postcode_space,`Average download speed (Mbit/s)`,`Minimum download speed (Mbit/s)`,`Maximum download speed (Mbit/s)`,County,Town)


# Writing new csv file for visualization
write_csv(broadband,"CleanedData/Broadband/CleanedBroadband.csv")
```

Figure 7

Cleaned Data

	postcode_space	Average download speed (Mbit/s)	Minimum download speed (Mbit/s)	Maximum download speed (Mbit/s)	County	Town
1	BA21 5BU	31.8	3.4	80.0	DORSET	YEOVIL
2	BA22 9RL	30.6	2.3	56.6	DORSET	YEOVIL
3	BA22 9RR	30.5	2.9	46.8	DORSET	YEOVIL
4	BA22 9RT	13.4	1.7	27.6	DORSET	YEOVIL
5	BA22 9RU	22.8	2.0	79.8	DORSET	YEOVIL
6	BA22 9RX	38.3	5.1	54.9	DORSET	YEOVIL
7	BA22 9RZ	34.8	4.9	80.0	DORSET	YEOVIL
8	BA22 9SA	44.7	3.4	80.0	DORSET	YEOVIL
9	BA22 9SB	20.4	2.3	79.9	DORSET	YEOVIL
10	BA22 9SE	48.1	5.3	63.9	DORSET	YEOVIL
11	BA22 9SF	45.7	2.8	79.9	DORSET	YEOVIL
12	BA22 9SG	37.5	5.3	72.1	DORSET	YEOVIL
13	BA22 9SN	36.9	2.3	73.9	DORSET	YEOVIL
14	BA22 9SR	31.0	3.2	75.9	DORSET	YEOVIL
15	BA22 9SW	2.6	0.6	3.4	DORSET	YEOVIL
16	BA22 9SZ	1.3	0.5	3.0	DORSET	YEOVIL
17	BA22 9UZ	19.8	12.8	40.0	DORSET	YEOVIL
18	BH12 4LS	20.2	3.2	79.9	DORSET	POOLE

Description

The broadband dataset was cleaned and prepared for analysis alongside the house price datasets for Devon and Dorset. Initially, the broadband speed data was processed by renaming columns for clarity and selecting only essential fields, including postcode_space, average_download_speed_mbps, maximum_download_speed_mbps, and minimum_download_speed_mbps. The cleaned broadband data was then merged with the house price datasets using left joins based on the PostCode column, integrating broadband speed information with property data for towns in Devon and Dorset. The resulting selected dataset, broadband, now combines house price details and broadband speed statistics, providing a comprehensive foundation for further analysis and regional insights.

Crime Dataset

Obtained Data

2021-11	2021-12	2022-01	2022-02
2022-03	2022-04	2022-05	2022-06
2022-07	2022-08	2022-09	2022-10
2022-11	2022-12	2023-01	2023-02
2023-03	2023-04	2023-05	2023-06
2023-07	2023-08	2023-09	2023-10
2023-11	2023-12		

Figure 8

	Crime.ID	Month	Reported.by	Falls.within	Longitude	Latitude	Location	LSOA.code	LSOA.na
1	fdbee9402642bf4f2e247e4f818e1e58a9ed366f9e6597a3afa...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-3.392966	54.87025	On or near Parking Area	E01019126	Allerdale
2	5266e6fd8c159a1d588831c164e8becce81ada588dff8d36e7...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-3.403712	54.84583	On or near B5300	E01019127	Allerdale
3	862bf5a1f900e5ffaccdc06b5cdd31114ebcbaae8120b5a6364...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.260856	52.24555	On or near Parking Area	W01000506	Ceredigic
4	8a120e84fde6e1e3077b5c35d81cac60703c916f2ff40c118c3f...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.195124	52.21079	On or near Maes Aeron	W01000524	Ceredigic
5	1e6731ada4b7e73603c10f8d6fa0a5348a7d48b0a915e91bd5...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.248988	52.24780	On or near Sports/Recreation Area	W01000540	Ceredigic
6	924ec2ddff1a2931419f6bc027d85f2a490895db1135f9fcc08f...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.248988	52.24780	On or near Sports/Recreation Area	W01000540	Ceredigic
7	f627ec16a691cb7afa4dba512053f1c8f96986c0a067943743f...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.248988	52.24780	On or near Sports/Recreation Area	W01000540	Ceredigic
8	0106937505c1073d2b6f66ea7aa315c8d47b619962de7a406...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.248988	52.24780	On or near Sports/Recreation Area	W01000540	Ceredigic
9	2929d9629599ae0cd6c9812649926b1973eacd34fa38b2d5a...	2021-11	Devon & Cornwall Police	Devon & Cornwall Police	-4.657686	52.11503	On or near Clos Y Gwyddil	W01000547	Ceredigic

Figure 9

Cleaning Code

```r
# Initialize an empty list to store CSV file paths
csv_files=list()

# Loop through each subfolder and collect all CSV file paths
for (folder in subfolders) {
  # List all CSV files in the current subfolder
  files_in_folder=list.files(folder, pattern = "\\.csv$", full.names = TRUE)

  # Add the CSV file paths to the list
  csv_files=c(csv_files, files_in_folder)
}

# Initialize an empty list to store data frames
data_list=list()

# Loop through each CSV file and read it into R
for (csv_file in csv_files) {
  # Read the CSV file into a data frame
  df=read.csv(csv_file)

  # Append the data frame to the list
  data_list=append(data_list, list(df))
}

# Merge all data frames into one data frame
merged_data=do.call(rbind, data_list)

# Save the merged data to a new CSV file
write.csv(merged_data, "CleanedData/Crime/CrimeData.csv", row.names = FALSE)
```

Clean Dataset

```r
# Initialize an empty list to store CSV file paths
csv_files=list()

# Loop through each subfolder and collect all CSV file paths
for (folder in subfolders) {
  # List all CSV files in the current subfolder
  files_in_folder=list.files(folder, pattern = "\\.csv$", full.names = TRUE)

  # Add the CSV file paths to the list
  csv_files=c(csv_files, files_in_folder)
}

# Initialize an empty list to store data frames
data_list=list()

# Loop through each CSV file and read it into R
for (csv_file in csv_files) {
  # Read the CSV file into a data frame
  df=read.csv(csv_file)

  # Append the data frame to the list
  data_list=append(data_list, list(df))
}

# Merge all data frames into one data frame
merged_data=do.call(rbind, data_list)

# Save the merged data to a new CSV file
write.csv(merged_data, "CleanedData/Crime/CrimeData.csv", row.names = FALSE)
```

Figure 10

Description

The crime dataset, initially separated into monthly files from 2021 to 2023, was merged into a single comprehensive dataset to facilitate analysis. This process began by systematically loading all monthly crime data files and appending them into a unified structure. Columns were standardized by renaming them to ensure consistency across all files. Duplicates were removed, and irrelevant columns were dropped to focus on essential information such as crime type, location, and date. The merged dataset was further cleaned by formatting date fields for uniformity, trimming whitespace, and ensuring proper data types for numeric and categorical fields. This consolidated and standardized crime dataset provides a clear and cohesive view of crime patterns over the three-year period, enabling more in-depth and accurate analysis.

School Dataset Cleaning

Obtained Dataset

	TOWN	PCODE	ATT8SCR	COUNTY	YEAR
1	Tiverton	EX16 5HB	7.9	DEVON	2021-2022
2	Bideford	EX39 5DW	36.2	DEVON	2021-2022
3	Ashburton	TQ13 7EW	52.7	DEVON	2021-2022
4	Exeter	EX4 5AD	41.9	DEVON	2021-2022
5	Axminster	EX13 5EA	42.0	DEVON	2021-2022
6	Exeter	EX4 1TA	4.6	DEVON	2021-2022
7	Exeter	EX4 1HS	6.6	DEVON	2021-2022
8	Bideford	EX39 3AR	43.5	DEVON	2021-2022
9	Tiverton	EX16 4DN	34.0	DEVON	2021-2022
10	Braunton	EX33 2BP	53.5	DEVON	2021-2022
11	Chulmleigh	EX18 7AA	57.7	DEVON	2021-2022
12	Exeter	EX5 3AJ	51.9	DEVON	2021-2022
13	Colyton	EX24 6HN	83.1	DEVON	2021-2022
14	Newton Abbot	TQ12 1PT	46.5	DEVON	2021-2022
15	Exeter	EX5 7EE	47.1	DEVON	2021-2022
16	Cullompton	EX15 1DX	46.8	DEVON	2021-2022
17	Dartmouth	TQ6 9HW	44.3	DEVON	2021-2022
18	Dawlish	EX7 0BY	45.1	DEVON	2021-2022
19	Exeter	EX2 4NS	21.6	DEVON	2021-2022
20	Exmouth	EX8 3AF	45.6	DEVON	2021-2022

Cleaning Code

```
#Load libraries

library(tidyverse)
library(dplyr)

#===============  Loading devon school data ==================
devon_school_21_22 <- read_csv("ObtainedData/2021-2022(Devon)/878_ks4final.csv")
devon_school_22_23 <- read_csv("ObtainedData/2022-2023(Devon)/878_ks4final.csv")

# ============== Loading dorset school data ===============================
dorset_school_21_22 <- read_csv("ObtainedData/2021-2022(Dorset)/838_ks4final.csv")
dorset_school_22_23 <- read_csv("ObtainedData/2022-2023(Dorset)/838_ks4final.csv")

# selecting only needed data and filtering the data
devon_school_21_22 <- devon_school_21_22 %>%
  mutate(ATT8SCR = as.numeric(ATT8SCR)) %>%
  filter(!is.na(ATT8SCR)) %>%
  mutate(COUNTY = "DEVON", YEAR = "2021-2022") %>%
  select(TOWN, PCODE, ATT8SCR, COUNTY, YEAR) %>%
  na.omit()

# selecting only needed data and filtering the data
devon_school_22_23 <- devon_school_22_23 %>%
  mutate(ATT8SCR = as.numeric(ATT8SCR)) %>%
  filter(!is.na(ATT8SCR)) %>%
  mutate(COUNTY = "DEVON", YEAR = "2022-2023") %>%
```

```
  na.omit()

# putting together in single data frame
devon_att_score <- rbind(devon_school_21_22,devon_school_22_23)

# selecting only needed data and filtering the data
dorset_school_21_22 <- dorset_school_21_22 %>%
  mutate(ATT8SCR = as.numeric(ATT8SCR)) %>%
  filter(!is.na(ATT8SCR)) %>%
  mutate(COUNTY = "DORSET", YEAR = "2021-2022") %>%
  select(TOWN, PCODE, ATT8SCR, COUNTY, YEAR) %>%
  na.omit()

# selecting only needed data and filtering the data
dorset_school_22_23 <- dorset_school_22_23 %>%
  mutate(ATT8SCR = as.numeric(ATT8SCR)) %>%
  filter(!is.na(ATT8SCR)) %>%
  mutate(COUNTY = "DORSET", YEAR = "2022-2023") %>%
  select(TOWN, PCODE, ATT8SCR, COUNTY, YEAR) %>%
  na.omit()

# putting together in single data frame
dorset_att_score <- rbind(dorset_school_21_22,dorset_school_22_23)

# merging the data of devon and dorset in single data frame
county_att8 <- rbind(devon_att_score,dorset_att_score)

# Saving the cleaned data in cleaned data folder
write_csv(county_att8,"CleanedData/School/CleanedSchool.csv")
```

Loading dorset school data

Clean Dataset

	TOWN	PCODE	ATT8SCR	COUNTY	YEAR
1	Weymouth	DT4 9BJ	41.4	DORSET	2021-2022
2	Portland	DT5 2NA	39.0	DORSET	2021-2022
3	Beaminster	DT8 3EP	47.2	DORSET	2021-2022
4	Blandford Forum	DT11 7SQ	49.7	DORSET	2021-2022
5	Blandford Forum	DT11 0PX	14.0	DORSET	2021-2022
6	Weymouth	DT4 9SY	43.2	DORSET	2021-2022
7	Blandford Forum	DT11 8LL	27.3	DORSET	2021-2022
8	Dorchester	DT2 8PX	30.5	DORSET	2021-2022
9	Ferndown	BH22 9EY	51.6	DORSET	2021-2022
10	Gillingham	SP8 4QP	52.2	DORSET	2021-2022
11	Sherborne	DT9 4EQ	53.5	DORSET	2021-2022
12	Sherborne	DT9 6EN	50.5	DORSET	2021-2022
13	Poole	BH16 6JD	51.5	DORSET	2021-2022
14	Blandford Forum	DT11 0BZ	18.9	DORSET	2021-2022
15	Wareham	BH20 4PF	49.5	DORSET	2021-2022
16	Wimborne	BH21 4DT	49.7	DORSET	2021-2022
17	Ringwood	BH24 2NN	5.4	DORSET	2021-2022
18	Shaftesbury	SP7 8ER	48.9	DORSET	2021-2022
19	Sherborne	DT9 3AP	33.4	DORSET	2021-2022
20	Sherborne	DT9 3QN	37.4	DORSET	2021-2022
21	Bridport	DT6 3DT	46.1	DORSET	2021-2022

Description

The dataset initially comprised a large number of columns, which were refined to focus on key details. Data from multiple academic years, stored in separate folders, was consolidated into a single unified dataset. To streamline the analysis, only essential columns—Postcode, Town, Attainment 8 score, and County—were retained. Relevant data was filtered, and data types were adjusted to ensure precision, with the Attainment 8 score converted to numeric format for accuracy. The cleaned dataset, featuring towns and their respective Attainment 8 scores, was saved in a "Cleaned Data" folder, ready for analysis and visualization.

Lsoa Dataset Cleaning

Obtained Data

pcd7	pcd8	pcds	dointr	doterm	usertype	oa11cd	lsoa11cd	msoa11cd	ladcd	lsoa11nm	msoa11nm	ladnm
AB1 0AA	AB1 0AA	AB1 0AA	198001	199606	0	S00090303	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AB	AB1 0AB	AB1 0AB	198001	199606	0	S00090303	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AD	AB1 0AD	AB1 0AD	198001	199606	0	S00090399	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AE	AB1 0AE	AB1 0AE	199402	199606	0	S00091322	S01006853	S02001296	S12000034	Dunecht, Durris and Drumoak - 01	Dunecht, Durris and Drumoak	Aberdeenshir
AB1 0AF	AB1 0AF	AB1 0AF	199012	199207	1	S00090299	S01006511	S02001236	S12000033	Culter - 06	Culter	Aberdeen City
AB1 0AG	AB1 0AG	AB1 0AG	199012	199207	1	S00090291	S01006506	S02001236	S12000033	Culter - 01	Culter	Aberdeen City
AB1 0AJ	AB1 0AJ	AB1 0AJ	198001	199606	0	S00090399	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AL	AB1 0AL	AB1 0AL	198001	199606	0	S00090381	S01006511	S02001236	S12000033	Culter - 06	Culter	Aberdeen City
AB1 0AN	AB1 0AN	AB1 0AN	198001	199606	1	S00090399	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AP	AB1 0AP	AB1 0AP	198001	199606	0	S00090399	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AQ	AB1 0AQ	AB1 0AQ	198001	199606	0	S00090399	S01006514	S02001237	S12000033	Cults, Bieldside and Milltimber West - 02	Cults, Bieldside and Milltimber Wes	Aberdeen City
AB1 0AR	AB1 0AR	AB1 0AR	198001	199606	0	S00092401	S01006853	S02001296	S12000034	Dunecht, Durris and Drumoak - 01	Dunecht, Durris and Drumoak	Aberdeenshir
AB1 0AS	AB1 0AS	AB1 0AS	198001	199606	0	S00092401	S01006853	S02001296	S12000034	Dunecht, Durris and Drumoak - 01	Dunecht, Durris and Drumoak	Aberdeenshir

Cleaning Code

```r
#load libraries
library(tidyverse)

# Loading data of lsoa code

lsoa=read_csv("ObtainedData/Postcode to LSOA.csv")

# renaming the column name and taking post code and lsoa code
l1=lsoa %>% select(lsoa11cd,pcds) %>% rename(lsoacode = lsoa11cd)

# Reading cleaned data generated from house price
postal_town=read_csv("CleanedData/Postal/PostalCodeTable.csv")

# performing joins to take common postcode and town ac=ssociated with lsoa code
lsoa_town=l1 %>%
  left_join(postal_town,by=c("pcds"="Postcode"))

# taking only data from DEVON and DORSET
lsoa_county=lsoa_town %>%
  filter((County=="DEVON"|County=="DORSET")&!is.na(lsoacode)) %>%
  group_by(Town,lsoacode)

# writing cleaned data in cleaned data foler
write_csv(lsoa_county,"CleanedData/Lsoa/CleanedLsoa.csv")
```

Clean Dataset

	lsoacode	pcds	Town	County
1	E01020497	BA21 5BU	YEOVIL	DORSET
2	E01020530	BA22 9RL	YEOVIL	DORSET
3	E01020530	BA22 9RR	YEOVIL	DORSET
4	E01020530	BA22 9RS	YEOVIL	DORSET
5	E01020530	BA22 9RT	YEOVIL	DORSET
6	E01020530	BA22 9RU	YEOVIL	DORSET
7	E01020530	BA22 9RX	YEOVIL	DORSET
8	E01020530	BA22 9RZ	YEOVIL	DORSET
9	E01020530	BA22 9SA	YEOVIL	DORSET
10	E01020530	BA22 9SB	YEOVIL	DORSET
11	E01020530	BA22 9SE	YEOVIL	DORSET
12	E01020530	BA22 9SF	YEOVIL	DORSET
13	E01020530	BA22 9SG	YEOVIL	DORSET
14	E01020530	BA22 9SN	YEOVIL	DORSET
15	E01020530	BA22 9SR	YEOVIL	DORSET
16	E01020530	BA22 9SW	YEOVIL	DORSET
17	E01020530	BA22 9SY	YEOVIL	DORSET
18	E01020530	BA22 9SZ	YEOVIL	DORSET
19	E01020497	BA22 9UZ	YEOVIL	DORSET
20	E01015443	BH12 4LS	POOLE	DORSET
21	E01020473	BH16 5BT	POOLE	DORSET

Description

While cleaning the LSOA data, the columns lsoa11cd and pcds were renamed to lsoacode and postcode, respectively, to enhance clarity and ensure consistency for further analysis. Renaming these columns helps make the data more intuitive by aligning the column names with their respective content, making it easier to interpret and work with in subsequent steps. By standardizing the naming conventions, the dataset becomes more user-friendly, reducing the chances of confusion or errors during later stages of analysis. This refinement ensures that the data is organized in a way that supports more efficient merging and querying when integrating it with other datasets, such as those for towns and counties.

Population Cleaning:

Obtained Data

	shortPostcode	Population2021	Population2022	Population2023
1	AL1	38647.9472	38864.8609	39082.9920
2	AL10	38348.8476	38564.0825	38780.5255
3	AL2	25214.1002	25355.6157	25497.9255
4	AL3	30403.4791	30574.1203	30745.7192
5	AL4	29128.0327	29291.5154	29455.9156
6	AL5	32302.7618	32484.0628	32666.3814
7	AL6	11987.4866	12054.7670	12122.4250
8	AL7	36201.7394	36404.9236	36609.2482
9	AL8	14155.9590	14235.4101	14315.3071
10	AL9	11119.0294	11181.4355	11244.1919
11	B1	8533.9538	8581.8511	8630.0172
12	B10	27675.2630	27830.5919	27986.7926
13	B11	47753.3950	48021.4135	48290.9363
14	B12	22170.7613	22295.1959	22420.3289
15	B13	39328.3989	39549.1317	39771.1033
16	B14	46374.3320	46634.6104	46896.3497
17	B15	19463.9095	19573.1518	19683.0071
18	B16	25391.4236	25533.9343	25677.2449
19	B17	27306.7295	27459.9900	27614.1107
20	B18	17393.7127	17491.3358	17589.5069
21	B19	21436.8990	21557.2147	21678.2057

Cleaning Code

```
library(tidyverse)
library(dplyr)
library(lubridate)

PopulationData = read_csv("ObtainedData/Population2011.csv")
PopulationData = PopulationData %>%  mutate(shortPostcode = str_trim(substring(Postcode, 1,4))) %>%
  group_by(shortPostcode) %>%
  summarise_at(vars(Population),list(Population2011 = sum)) %>%
  mutate(Population2012= (1.00695353132322269 * Population2011)) %>%
  mutate(Population2013= (1.00669740535540783 * Population2012)) %>%
  mutate(Population2014= (1.00736463978721671 * Population2013)) %>%
  mutate(Population2015= (1.00792367505802859 * Population2014)) %>%
  mutate(Population2016= (1.00757874492811929 * Population2015)) %>%
  mutate(Population2017= (1.00679374473924223 * Population2016)) %>%
  mutate(Population2018= (1.00605929132212552 * Population2017)) %>%
  mutate(Population2019= (1.00561255390388033 * Population2018)) %>%
  mutate(Population2020= (1.00561255390388033 * Population2019)) %>%
  mutate(Population2021= (1.00561255390388033 * Population2020)) %>%
  mutate(Population2022= (1.00561255390388033 * Population2021)) %>%
  mutate(Population2023= (1.00561255390388033 * Population2022)) %>%
  #Add other year use samne formula 1.00561255390388033 for every year to 2022
  select(shortPostcode,Population2021,Population2022,Population2023)

HousePrices=read_csv("CleanedData/Postal/PostalCodeTable.csv")
#Towns code should be done because population data needs to be added in Housprice Cleaned Data

Towns = HousePrices %>%
  mutate(shortPostcode = str_trim(substring(Postcode, 1,4))) %>%
  left_join(PopulationData,by = "shortPostcode") %>%
  select(shortPostcode,Postcode,Town,County,Population2021,Population2022,Population2023) %>%
  group_by(shortPostcode) %>%
  filter(row_number()==1) %>%
  arrange(County)

Population=Towns %>%
  select(Postcode,Town,Population2021,Population2022,Population2023)

write_csv(Population,"CleanedData/Population/CleanedPopulation.csv")
```

Clean Data

	shortPostcode	Postcode	Town	Population2021	Population2022	Population2023
1	EX18	EX18 7BE	CHULMLEIGH	3329.620	3348.308	3367.100
2	PL19	PL19 9AG	TAVISTOCK	17543.262	17641.725	17740.740
3	EX14	EX14 3NH	HONITON	22098.123	22222.150	22346.873
4	EX4	EX4 2LT	EXETER	58443.003	58771.017	59100.873
5	TQ13	TQ13 9AL	NEWTON ABBOT	24712.040	24850.738	24990.214
6	EX2	EX2 6DS	EXETER	43474.134	43718.135	43963.505
7	EX39	EX39 5DP	BIDEFORD	39698.001	39920.808	40144.865
8	TQ10	TQ10 9EX	SOUTH BRENT	4303.830	4327.986	4352.277
9	TQ9	TQ9 7NU	TOTNES	18436.289	18539.763	1 4352.277
10	EX17	EX17 3LA	CREDITON	20871.814	20988.958	21106.760
11	TQ12	TQ12 6QN	NEWTON ABBOT	56608.881	56926.601	57246.105
12	EX22	EX22 7AW	HOLSWORTHY	11477.949	11542.370	11607.152
13	EX13	EX13 5AN	AXMINSTER	12101.785	12169.707	12238.010
14	EX1	EX1 2RH	EXETER	22989.012	23118.040	23247.791
15	EX32	EX32 9HP	BARNSTAPLE	25399.969	25542.528	25685.887
16	EX20	EX20 1QS	OKEHAMPTON	21327.941	21447.645	21568.022
17	EX15	EX15 1FN	CULLOMPTON	22766.824	22894.604	23023.101
18	EX7	EX7 0PQ	DAWLISH	13280.024	13354.559	13429.512

Description

This code integrates population data with town information, focusing on the years 2021-2023. It begins by estimating population for each year (2012-2023) based on a growth rate formula applied to 2011 data. The population is grouped by the first four characters of the postcode. This data is then merged with house price data, linking the two datasets by the shortPostcode field. The final dataset, containing postcode, town, and population estimates for 2021-2023, is saved for further analysis. The goal is to combine demographic and geographical data for town-level insights and comparisons.

Exploratory Data Analysis (EDA

Exploratory Data Analysis (EDA) is the process of analyzing and summarizing datasets to uncover patterns, spot anomalies, test hypotheses, and check assumptions using visualizations and statistical methods. It serves as a critical initial step in the data analysis process, providing a deeper understanding of the data's structure, distribution, and relationships between variables. EDA is essential for identifying data quality issues, such as missing or inconsistent values, and for guiding data preprocessing and feature engineering decisions. By leveraging EDA, analysts can derive meaningful insights, ensure the dataset is ready for advanced modeling, and build a solid foundation for data-driven decision-making.

House Dataset Visualization

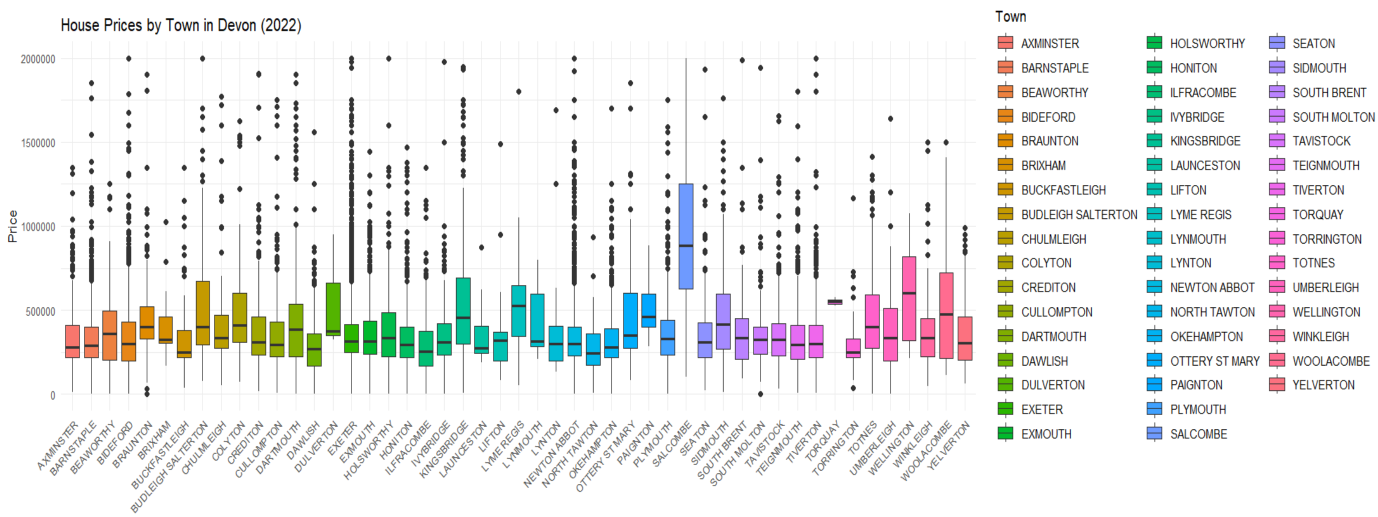


Figure 11: Boxplot House Price Devon

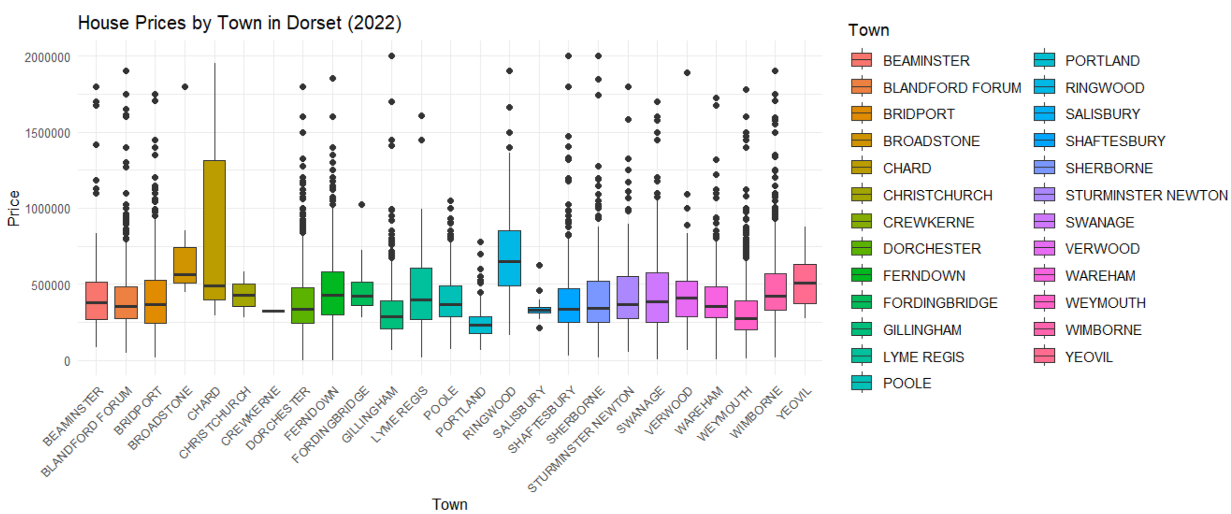


Figure 12 : Boxplot House Price Dorset

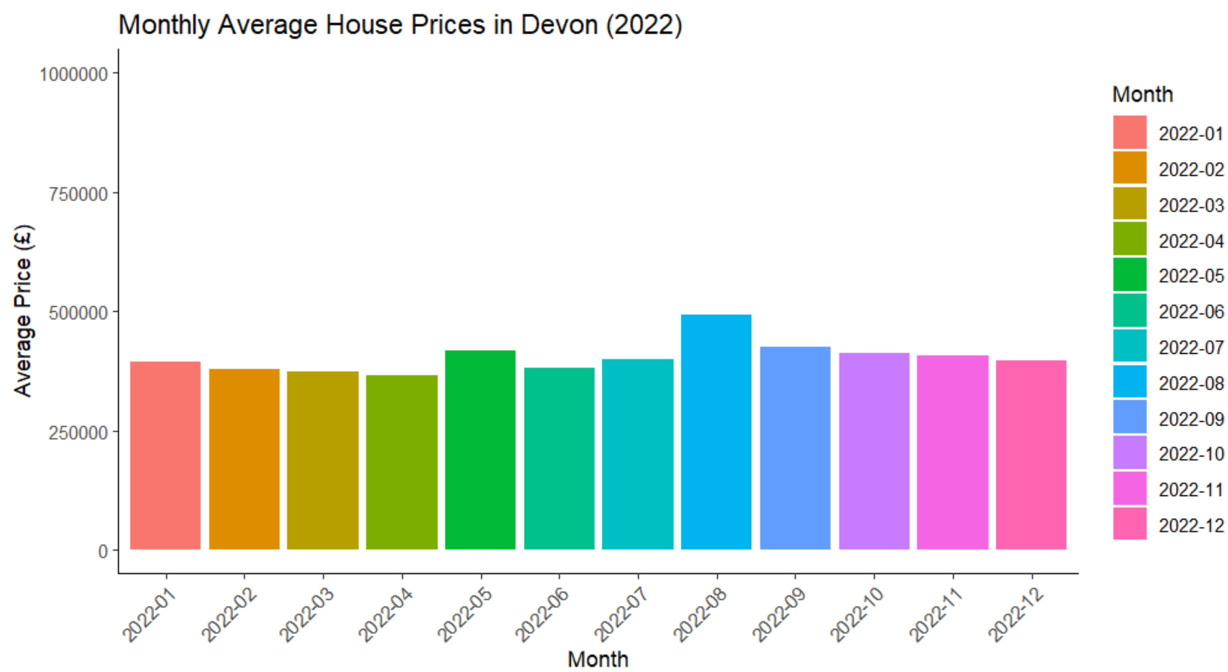


Figure 13 : Bar Chart Monthly Average Devon

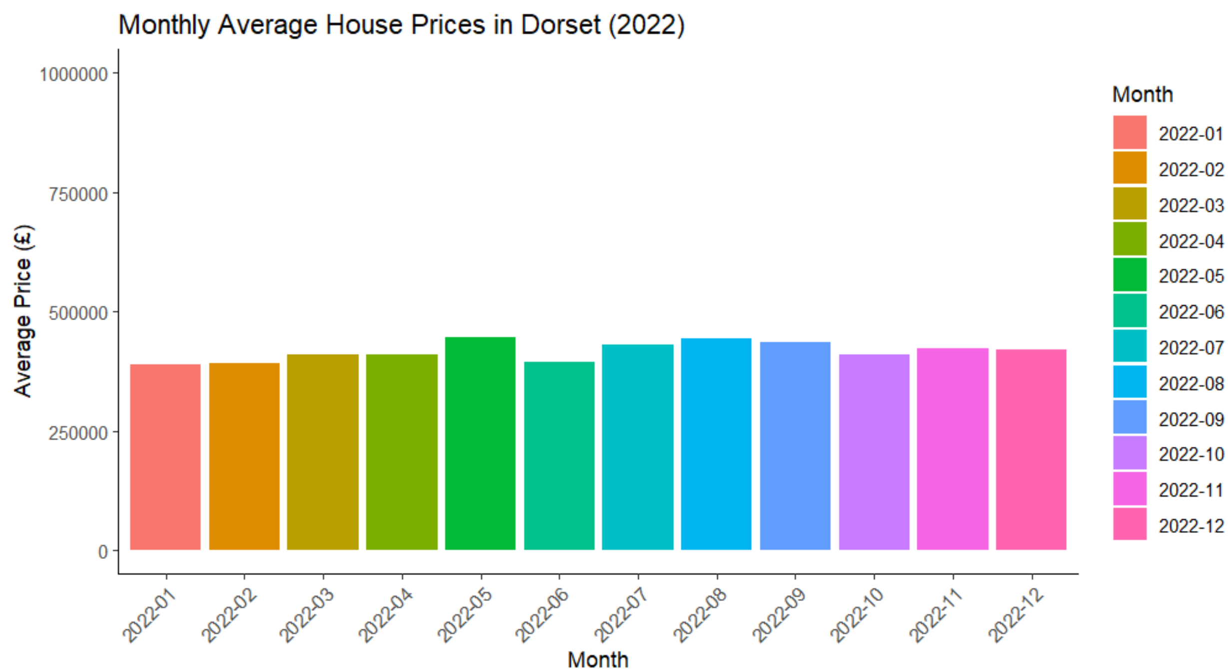


Figure 14 : Bar Chart Monthly Average Dorset

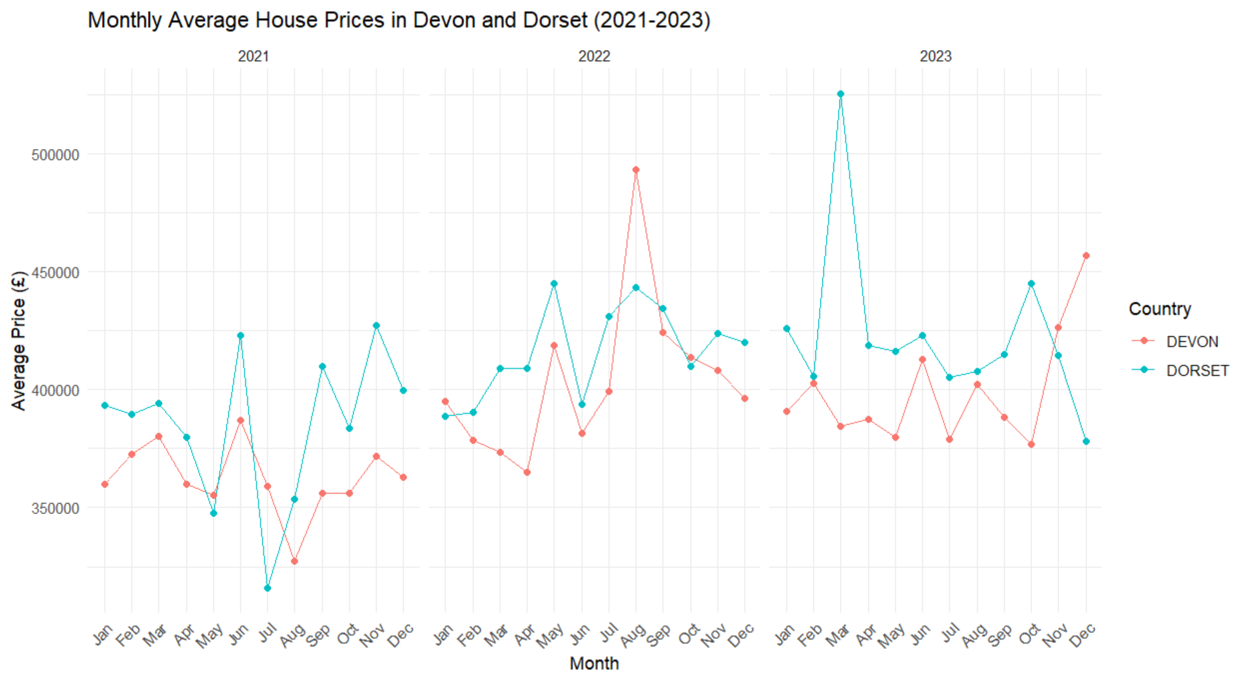


Figure 15 : Line Chart House Price (2021 -2023)

Code for Visualization

```r
library(tidyverse)
library(dplyr)
library(lubridate)

Dorset=read_csv("CleanedData/House/DorsetHousePrice.csv")
Devon=read_csv("CleanedData/House/DevonHousePrice.csv")

n=Dorset %>%
  summarize(n=n_distinct(Town))


# Boxplot for Average Dorset House price for year 2022 according to Town
Dorset %>%
  filter(year(DateOfTransaction) == 2022) %>%
  select(Town, Price) %>%
  group_by(Town) %>%
  ggplot(aes(x = Town, y = Price, fill = Town)) +
  geom_boxplot() +
  theme_minimal() +
  labs(
    title = "House Prices by Town in Dorset (2022)",
    x = "Town",
    y = "Price"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    axis.text = element_text(size = 8),
    axis.title = element_text(size = 10),
    plot.title = element_text(size = 12)
  )+ylim(0,2000000)

# Boxplot for Average Devon House price for year 2022 according to Town
Devon %>%
  filter(year(DateOfTransaction) == 2022) %>%
  select(Town, Price) %>%
  group_by(Town) %>%
  ggplot(aes(x = Town, y = Price, fill = Town)) +
  geom_boxplot() +
  theme_minimal() +
  labs(
    title = "House Prices by Town in Devon (2022)",
    x = "Town"
```

```r
# Barchart for avg house price in towns of Devon for year 2022

combined_house_dataset <- rbind(Dorset , Devon)

# Calculate Monthly Average Prices - Devon and Dorset (2022)
monthly_avg_prices <- combined_house_dataset %>%
  filter(
    County %in% c("DEVON", "DORSET"),
    format(as.Date(DateOfTransaction), "%Y") == "2022",
    !is.na(Price)
  ) %>%
  mutate(Month = format(as.Date(DateOfTransaction), "%Y-%m")) %>%
  group_by(County, Month) %>%
  summarise(average_price = mean(Price, na.rm = TRUE), .groups = "drop")



ggplot(monthly_avg_prices %>% filter(County == "DEVON"), aes(x = Month, y = average_price, fill = Month)) +
  geom_col() +
  labs(
    title = "Monthly Average House Prices in Devon (2022)",
    x = "Month",
    y = "Average Price (£)"
  ) +
  scale_y_continuous(limits = c(0, 1e6)) +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))



# Dorset
ggplot(monthly_avg_prices %>% filter(County == "DORSET"), aes(x = Month, y = average_price, fill = Month)) +
  geom_col() +
  labs(
    title = "Monthly Average House Prices in Dorset (2022)",
    x = "Month",
    y = "Average Price (£)"
  ) +
  scale_y_continuous(limits = c(0, 1e6)) +
  theme_classic() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```r
# Combined Analysis for Devon and Dorset (2021-2023)
filtered_dataset <- combined_house_dataset %>%
  filter(
    County %in% c("DEVON", "DORSET"),
    format(as.Date(DateOfTransaction), "%Y") %in% c("2021", "2022", "2023")
  ) %>%
  mutate(
    Year = format(as.Date(DateOfTransaction), "%Y"),
    Month = format(as.Date(DateOfTransaction), "%b")
  ) %>%
  group_by(County, Year, Month) %>%
  summarise(average_price = mean(Price, na.rm = TRUE), .groups = "drop")

# Ensure months are ordered correctly
filtered_dataset$Month <- factor(filtered_dataset$Month, levels = month.abb)

# Line Chart for Monthly Average Prices
ggplot(filtered_dataset, aes(x = Month, y = average_price, color = County, group = interaction(County, Year))) +
  geom_line() +
  geom_point() +
  facet_wrap(~Year) +
  labs(
    title = "Monthly Average House Prices in Devon and Dorset (2021-2023)",
    x = "Month",
    y = "Average Price (£)",
    color = "Country"
  ) +
  scale_x_discrete(limits = month.abb) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Description

The boxplot illustrating average house prices in 2023, categorized by towns in Devon and Dorset, offers a comparative view of price distributions. The median line within each box represents the typical house price for a town, with higher medians indicating generally more expensive housing. The interquartile range (IQR), depicted as the box, captures the central 50% of the data, with wider boxes suggesting greater variation in prices. Whiskers extend to values within 1.5 times the IQR, while points beyond the whiskers represent outliers, highlighting towns with exceptionally high or low prices. By examining medians, IQRs, and outliers, we can pinpoint towns with higher costs, greater price variability, and any anomalies, while also comparing the overall price patterns between Devon and Dorset.

The bar chart of average house prices by town provides a straightforward comparison, with bar heights indicating the average price in each town. This allows for easy identification of the most and least expensive towns. Using distinct colors or grouping for towns in Devon and Dorset enhances clarity and facilitates intuitive comparisons.

The line chart showing monthly average house prices from 2021 to 2023 for Devon and Dorset illustrates trends over time. The x-axis represents months, while the y-axis indicates average prices, with separate lines for each county. The slope of the lines reflects the rate of price changes, where steeper slopes indicate faster increases or decreases. This visualization highlights long-term trends, significant price fluctuations, and differences in the housing markets of the two counties.

Broadband Dataset Visualization

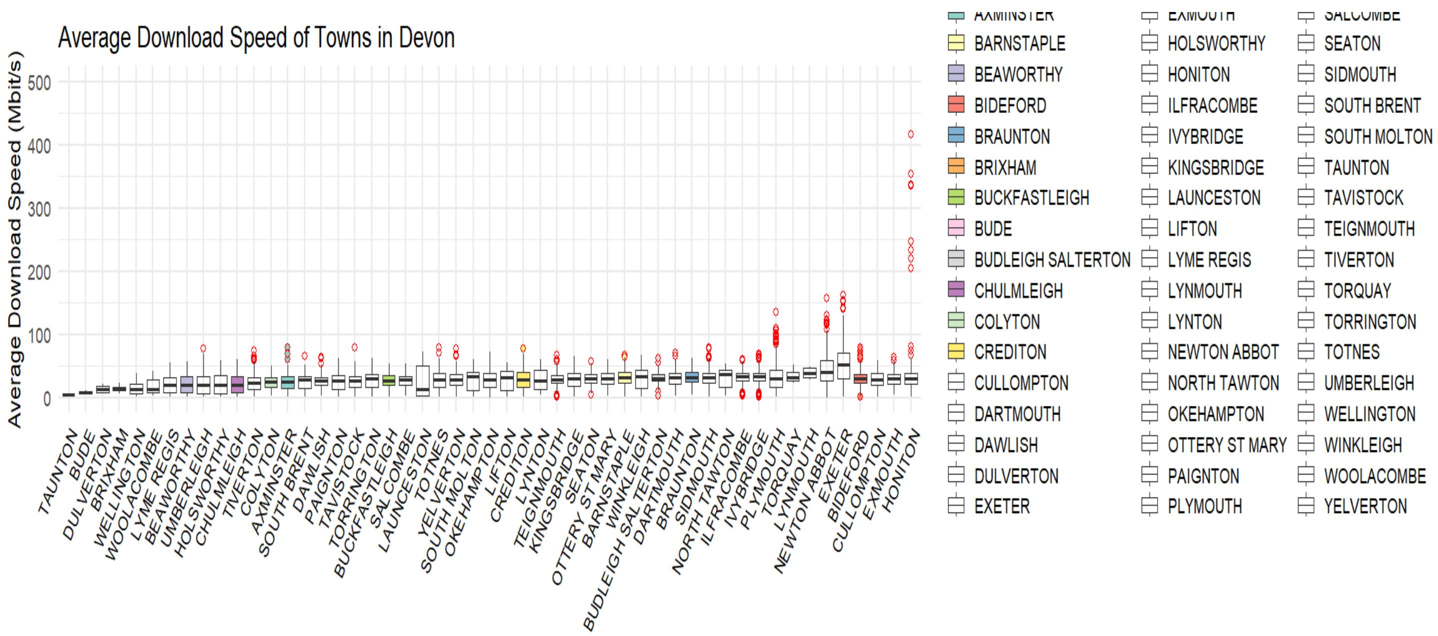


Figure 16 : Boxplot for Average Download speed Devon

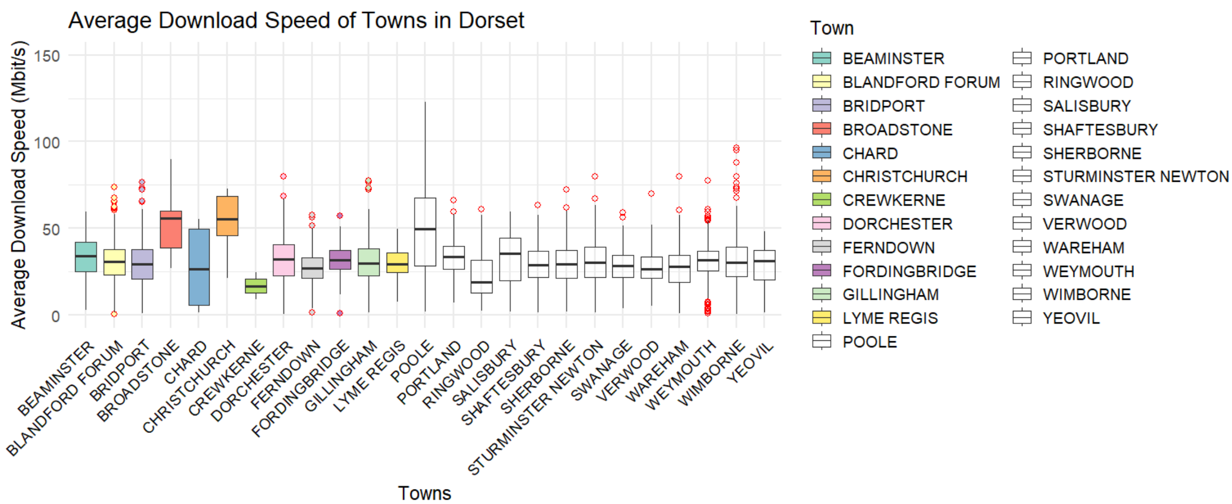


Figure 17 : Boxplot for Average Download speed Dorset

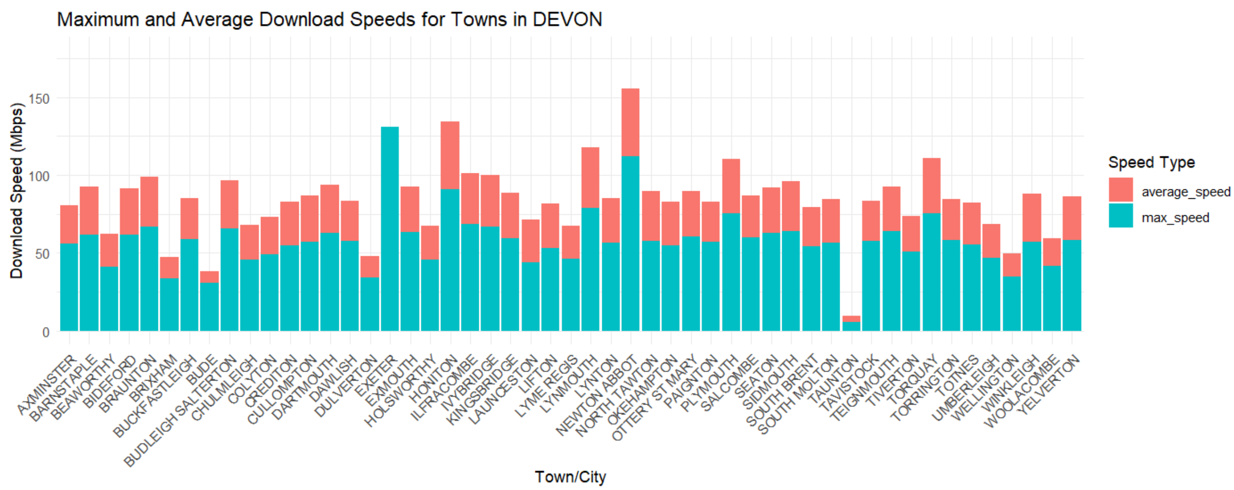


Figure 18 : Barchart Max and Avg for Devon

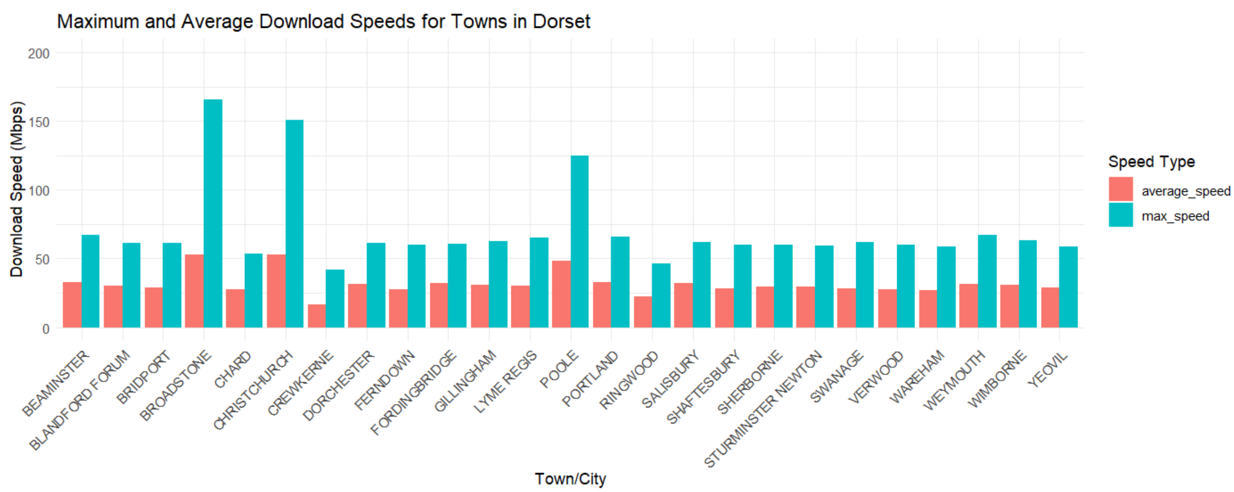


Figure 19 : Barchart Max and Avg for Dorset

Code for Visualization

```r
#load libraries

library(tidyverse)
library(dplyr)
library(lubridate)

broadband=read_csv("CleanedData/Broadband/CleanedBroadband.csv")

#Boxplot for average download speed of towns in DEVON

broadband %>%
  filter(County == "DEVON") %>%
  group_by(Town) %>%
  ggplot(aes(x = reorder(Town, `Average download speed (Mbit/s)`), y = `Average download speed (Mbit/s)`, fill = Town)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 21, outlier.size = 2) +
  scale_fill_brewer(palette = "Set3") + # Use a color palette for better aesthetics
  labs(
    title = "Average Download Speed of Towns in Devon",
    x = "Towns",
    y = "Average Download Speed (Mbit/s)",
    fill = "Town"
  ) +
  theme_minimal(base_size = 14) + # Use a minimal theme with increased base font size
  theme(
    axis.text.x = element_text(angle = 60, hjust = 1, vjust = 1, color = "black")
  ) +
  ylim(0, 500)

#Boxplot for average download speed of towns in DORSET
broadband %>%
  filter(County == "DORSET") %>%
  group_by(Town) %>%
  ggplot(aes(x = Town, y = `Average download speed (Mbit/s)`, fill = Town)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 21, outlier.size = 2) +
  scale_fill_brewer(palette = "Set3") + # Use a color palette for better aesthetics
  labs(
    title = "Average Download Speed of Towns in Dorset",
    x = "Towns",
    y = "Average Download Speed (Mbit/s)",
    fill = "Town"
  ) +
  theme_minimal(base_size = 14) + # Use a minimal theme with increased base font size
  theme(
```

```r
 #==================================== barchart ============================
#===================== dorset data ===============================
ggplot(dorset_data, aes(x = Town, y = Speed, fill = SpeedType)) +
  geom_col(position = "dodge") +
  labs(
    title = "Maximum and Average Download Speeds for Towns in Dorset",
    x = "Town/City",
    y = "Download Speed (Mbps)",
    fill = "Speed Type"
  ) +
  scale_y_continuous(
    breaks = seq(0, 300, by = 30),
    limits = c(0, 300)
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))  + ylim(0 , 200)
#============================================================================
devon_data <- broadband %>%
  filter(County == "DEVON") %>% # Adjust column name for county if different
  group_by(Town) %>%
  summarise(
    average_speed  = mean(`Average download speed (Mbit/s)` , na.rm = TRUE) ,
    max_speed = mean(`Maximum download speed (Mbit/s)` , na.rm = TRUE)
  ) %>%
  pivot_longer(cols = c(average_speed, max_speed), names_to = "SpeedType", values_to = "Speed")
#======================== Barchart ===================================
#======================== Devon ======================================

ggplot(devon_data, aes(x = Town, y = Speed, fill = SpeedType)) +
  geom_col() +
  labs(
    title = "Maximum and Average Download Speeds for Towns in DEVON",
    x = "Town/City",
    y = "Download Speed (Mbps)",
    fill = "Speed Type"
  ) +
  scale_y_continuous(
    limits = c(0, 180)
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Description:

The boxplot for average download speeds, grouped by towns in Devon and Dorset, provides a detailed look at the distribution of speeds. The median line within each box represents the typical download speed for a town, with higher medians indicating faster connectivity. The interquartile range (IQR), shown as the box, covers the middle 50% of the data, where wider boxes signal greater variation in speeds. Whiskers extend to values within 1.5 times the IQR, while data points outside this range are outliers, highlighting towns with exceptionally high or low speeds. Analyzing medians, IQRs, and outliers helps pinpoint towns with superior speeds, areas with significant variability, and locations that may face connectivity challenges, while also uncovering differences in speed patterns between Devon and Dorset.

The bar chart comparing average and maximum download speeds by town offers a clear side-by-side view of both metrics. Each bar's height represents the speed, enabling easy identification of towns with the fastest and slowest connections. Including maximum speeds alongside averages highlights the potential for higher speeds even in towns where the average is lower. This visualization makes it simple to compare towns, uncover discrepancies between typical and peak speeds, and analyze trends or relationships between these metrics across Devon and Dorset.

School Dataset Visualization

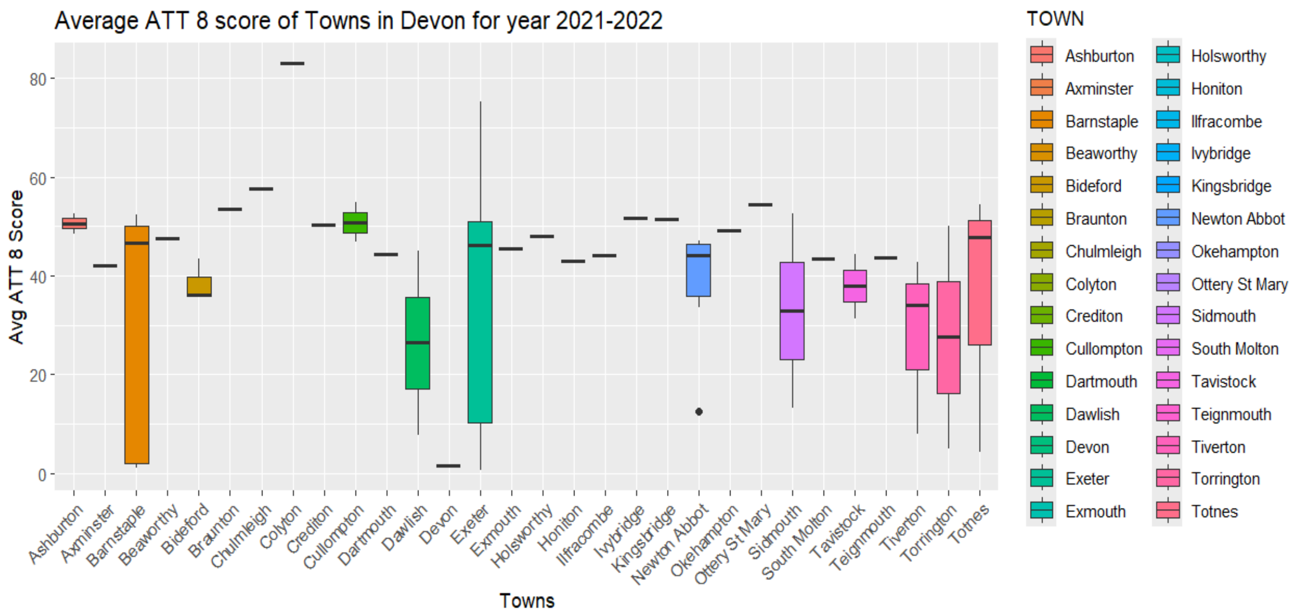


Figure 20 : Bxplots Average ATT8 Score Devon

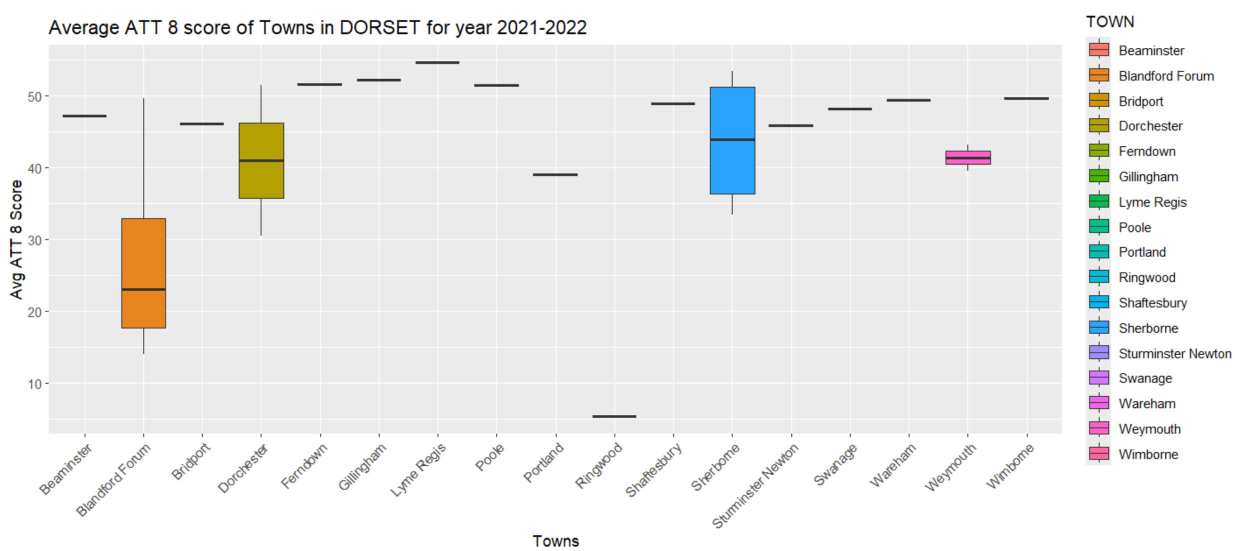


Figure 21 : Boxplots Average ATT8 Score Dorset

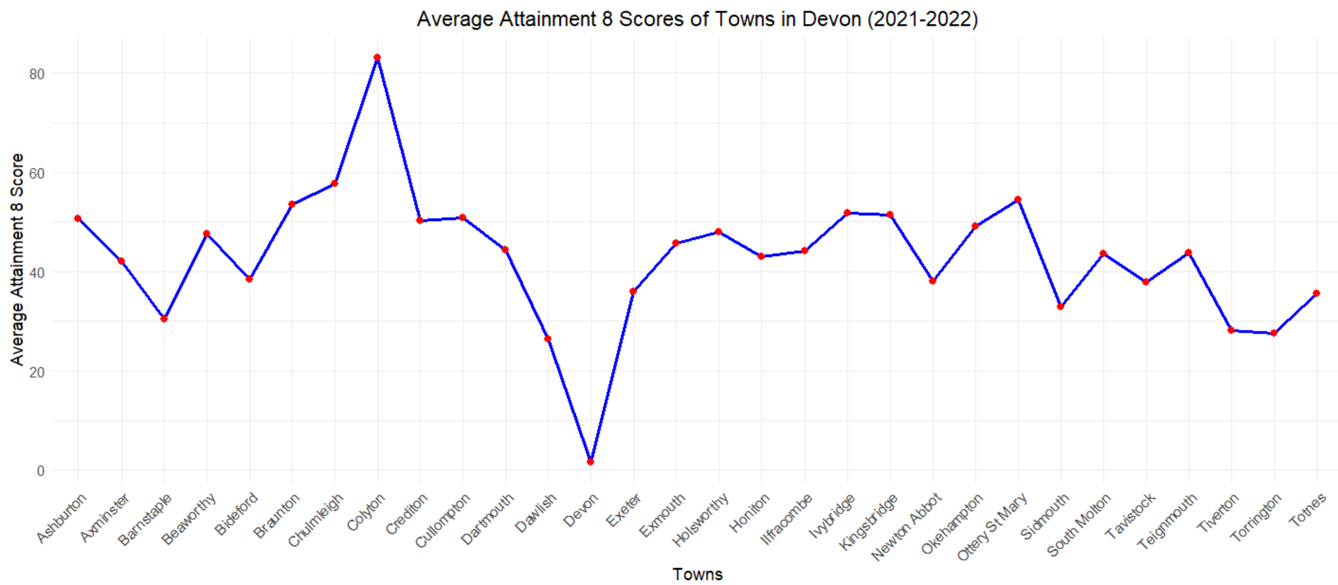


Figure 22 : Line Chart ATT8 Score Devon

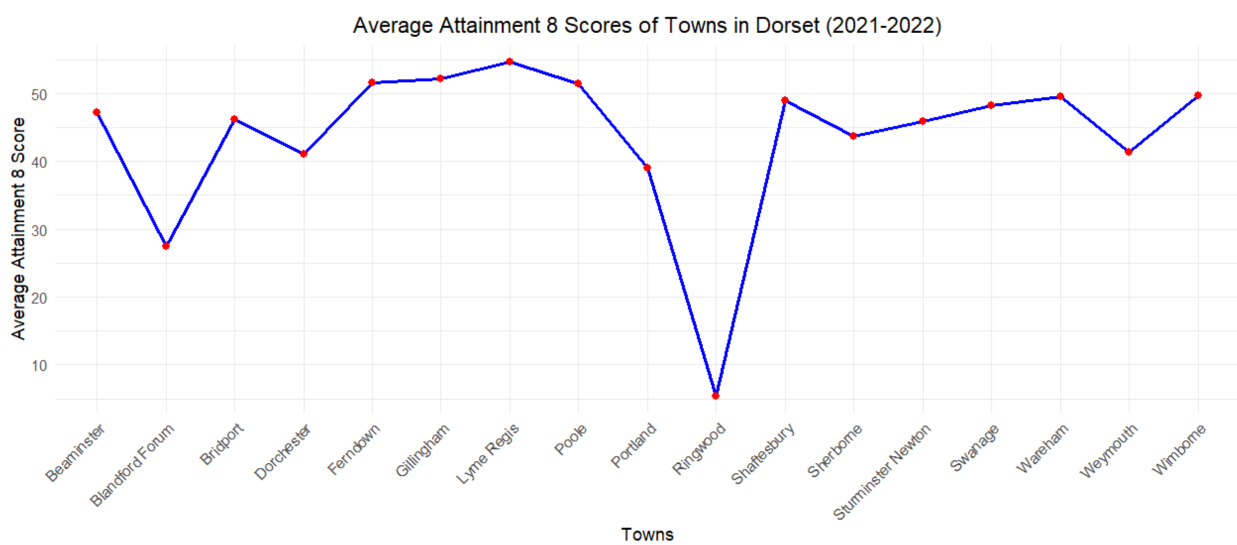


Figure 23 : Line Chart Att. Score Dorset

Code for Visualization

```r
#load libraries

library(tidyverse)
library(dplyr)

school <- read_csv("CleanedData/School/CleanedSchool.csv")

devon_school=school %>%
  filter(COUNTY=="DEVON",YEAR=="2021-2022")

devon_school %>%
  ggplot(aes(x=TOWN,y=ATT8SCR,fill = TOWN))+
  geom_boxplot()+
  labs(
    title = "Average ATT 8 score of Towns in Devon for year 2021-2022",
    x="Towns",
    y="Avg ATT 8 Score",
  )+
  theme(
    axis.text.x=element_text(angle = 45,hjust = 1)
  )

dorset_school <- school %>%
  filter(COUNTY =="DORSET",YEAR=="2021-2022")

dorset_school %>%
  ggplot(aes(x=TOWN,y=ATT8SCR,fill = TOWN))+geom_boxplot()+labs(
    title = "Average ATT 8 score of Towns in DORSET for year 2021-2022",
    x="Towns",
    y="Avg ATT 8 Score",
  )+
  theme(
    axis.text.x=element_text(angle = 45,hjust = 1)
  )
```

```r
#====================Line chart ===================
#================== Devon School ==================
devon_school_avg <- devon_school %>%
  group_by(TOWN) %>%
  summarize(Avg_ATT8SCR = mean(ATT8SCR))

devon_school_avg %>%
  ggplot(aes(x = TOWN, y = Avg_ATT8SCR, group = 1)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(
    title = "Average Attainment 8 Scores of Towns in Devon (2021-2022)",
    x = "Towns",
    y = "Average Attainment 8 Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
  )

#===================== Line Chart =======================
#===================== Dorset School =====================
dorset_school_avg <- dorset_school %>%
  group_by(TOWN) %>%
  summarize(Avg_ATT8SCR = mean(ATT8SCR))

dorset_school_avg %>%
  ggplot(aes(x = TOWN, y = Avg_ATT8SCR, group = 1)) +
  geom_line(color = "blue", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(
    title = "Average Attainment 8 Scores of Towns in Dorset (2021-2022)",
    x = "Towns",
    y = "Average Attainment 8 Score"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    plot.title = element_text(hjust = 0.5)
```

Description

The boxplot for average Attainment 8 scores (2022-2023) by town in Devon and Dorset illustrates the distribution of academic performance across towns. The median, shown as the central line in each box, represents the typical score for a town, with higher medians indicating stronger performance. The interquartile range (IQR), represented by the box, captures the central 50% of scores, where wider boxes indicate greater variation in performance. Whiskers extend to scores within 1.5 times the IQR, while points beyond the whiskers mark outliers, highlighting towns with unusually high or low scores. By examining medians, variability, and outliers, we can identify towns with better overall performance, those with more variability in scores, and towns that deviate significantly from the norm. Additionally, this visualization reveals differences in the distribution of scores between Devon and Dorset.

The line chart for Attainment 8 scores by town offers a comparison of performance trends over academic years. The x-axis represents the years, while the y-axis shows the average scores, with each line corresponding to a town. Upward slopes indicate improvements in academic performance, while downward slopes suggest a decline. This chart provides insights into how scores have evolved over time, identifies towns showing consistent improvement or decline, and highlights notable differences in performance trends between towns in Devon and Dorset.

Crime Dataset Visualization

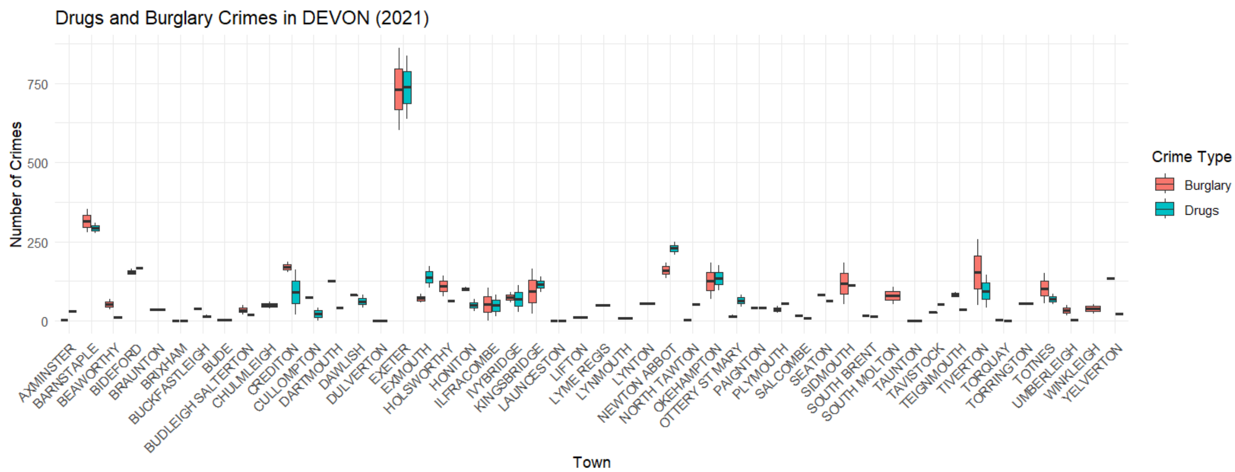


Figure 24 : Boxplot Drugs and Burglary (Devon)

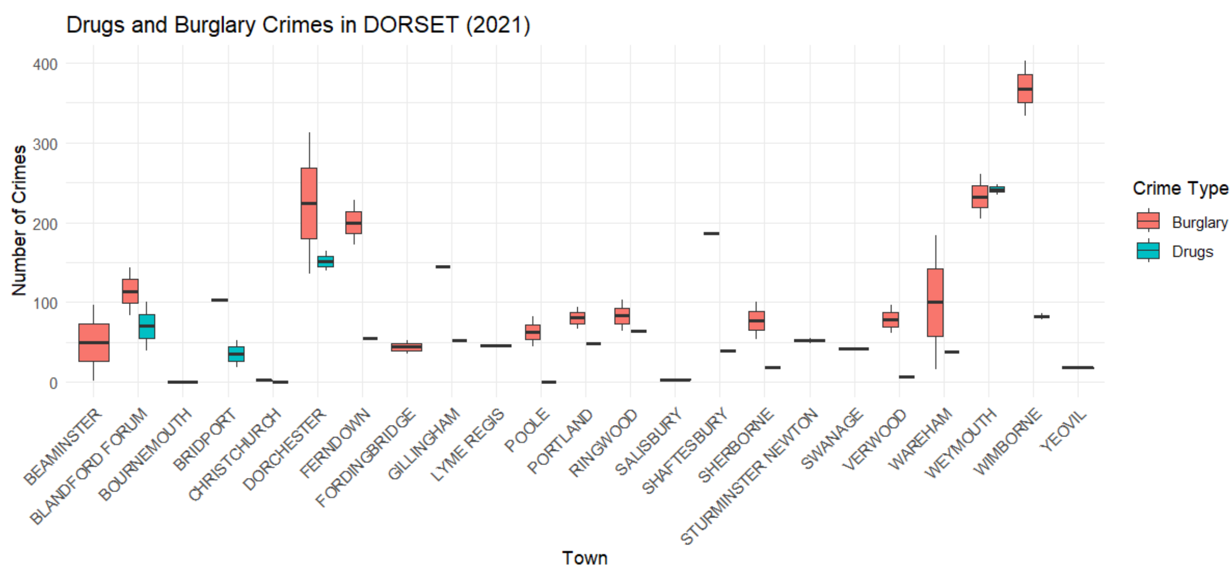


Figure 25 : Boxplot Drug and Burglary (Dorset)

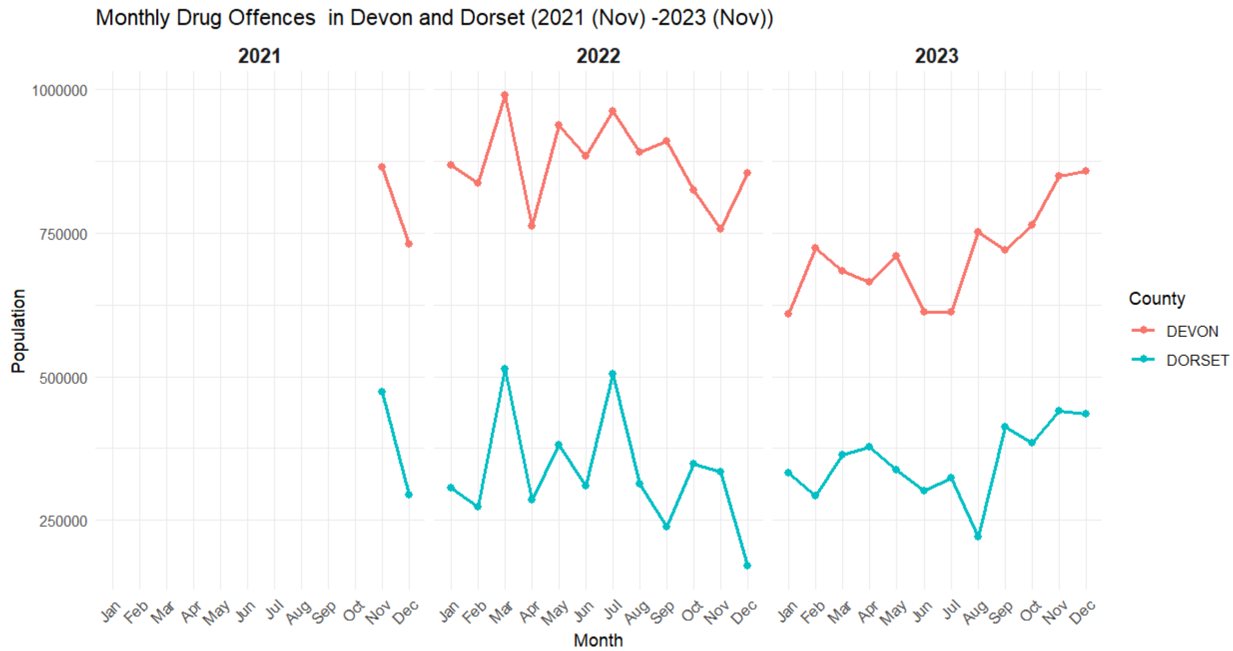


Figure 26 : Line chart Drug Offences (2021 -2023)

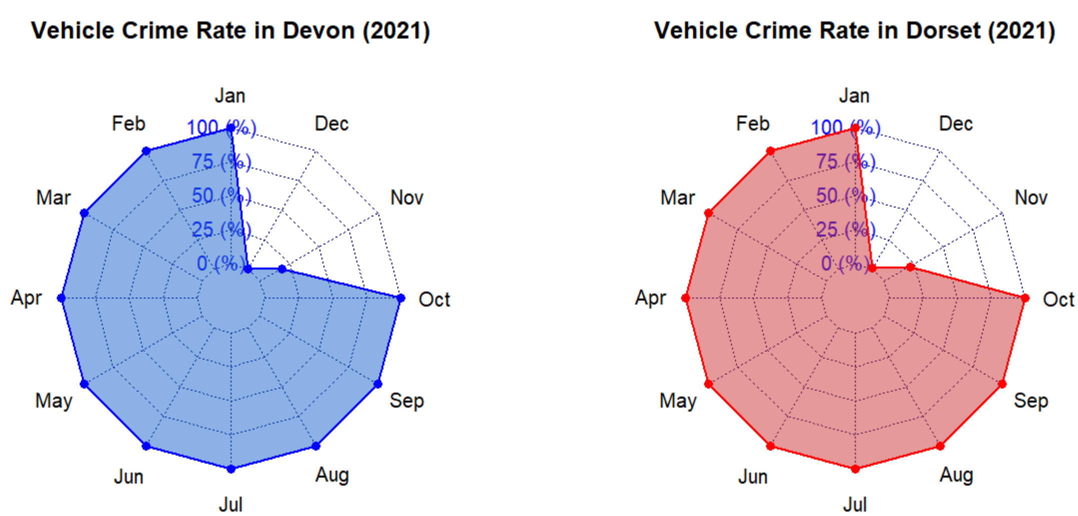


Figure 27 : Radar Chart Vechile Crime Rate (2021)

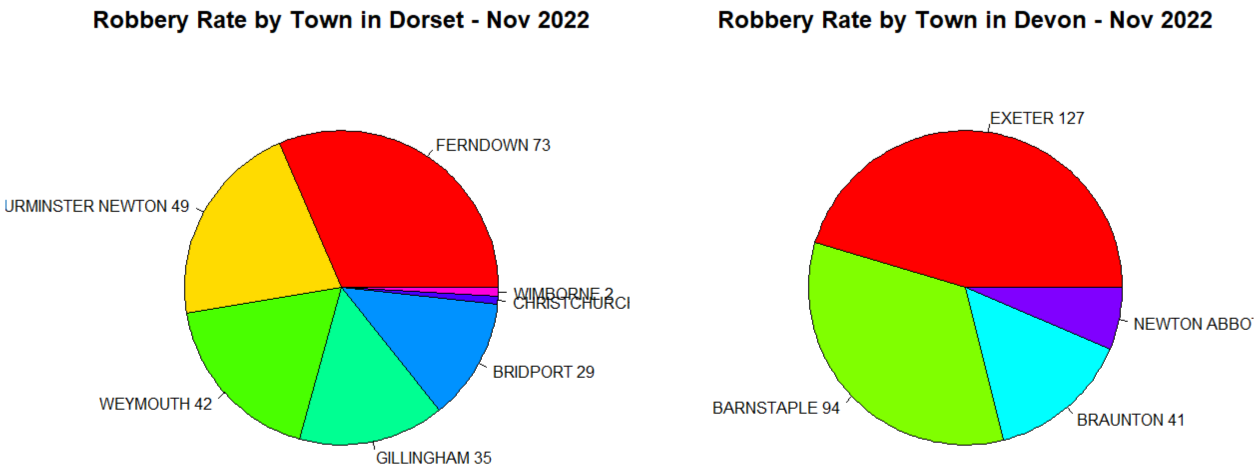


Figure 28 : Robbery Rate by Dorset (2022)

Code for Visualization

```r
#========================================== Boxplot ===============================
#==================================== Devon Drugs and Bruglary ========================

ggplot(Devon_crime_boxplot, aes(x = Town, y = CrimeCount, fill = CrimeType)) +
  geom_boxplot() +
  labs(
    title = "Drugs and Burglary Crimes in DEVON (2021)",
    x = "Town",
    y = "Number of Crimes",
    fill = "Crime Type"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))



#===================================================Boxplot ==============
#=========================Dorsest Drugs and Burglary ==========

Dorset_crime_boxplot <- crime_merged %>%
  filter(year(Month) == 2021,
         County == "DORSET",
         Crime.type %in% c("Drugs", "Burglary"))


ggplot(Dorset_crime_boxplot, aes(x = Town, y = CrimeCount, fill = Crime.type)) +
  geom_boxplot() +
  labs(
    title = "Drugs and Burglary Crimes in DORSET (2021)",
    x = "Town",
    y = "Number of Crimes",
    fill = "Crime Type"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```r
#============================Radar Chart calculation ==================================
crime_2021 <- crime_merged %>%
  filter(year(Month)==2021 & Crime.type == "Vehicle crime")

devon_data=crime_2021 %>%
  filter(County == "DEVON") %>%
  group_by(Month) %>%
  summarize(CrimeCount = sum(CrimeCount))

dorset_data=crime_2021 %>%
  filter(County == "DORSET") %>%
  group_by(Month) %>%
  summarize(CrimeCount = sum(CrimeCount))

#================================= Radar chart Visualization ==================
radar_data_prep =function(data) {
  # Ensure Month is in "Jan", "Feb" format
  data = data %>%
    mutate(Month = format(as.Date(paste0(Month, "-01")), "%b")) %>%
    complete(Month = month.abb, fill = list(CrimeCount = 0)) %>%
    arrange(match(Month, month.abb)) %>%
    select(CrimeCount)

  # Add min and max rows
  radar_data= rbind(
    rep(0, 12),                    # Min values
    rep(max(data$CrimeCount) + 10, 12), # Max values
    as.numeric(data$CrimeCount)    # Actual values
  )

  # Add column names
  colnames(radar_data)= month.abb
  return(radar_data)
}

# Prepare data for radar chart
devon_radar=radar_data_prep(devon_data)
dorset_radar=radar_data_prep(dorset_data)
devon_radar=as.data.frame(devon_radar)
dorset_radar=as.data.frame(dorset_radar)
```

```r
# Plot radar chart for Devon
par(mfrow = c(1, 2))  # Side-by-side plots
radarchart(devon_radar, axistype = 1,
           pcol = "blue", pfcol = rgb(0.1, 0.4, 0.8, 0.5), plwd = 2,
           title = "Vehicle Crime Rate in Devon (2021)")

# Plot radar chart for Dorset
radarchart(dorset_radar, axistype = 1,
           pcol = "red", pfcol = rgb(0.8, 0.2, 0.2, 0.5), plwd = 2,
           title = "Vehicle Crime Rate in Dorset (2021)")



#Pie chart
crime_nov_2022 <- crime_merged %>%
  filter(format(Month, "%Y-%m") == "2022-11" & Crime.type == "Robbery")

devon_robbery_town=crime_nov_2022 %>%
  filter(County == "DEVON") %>%
  group_by(Town) %>%
  summarize(CrimeCount = sum(CrimeCount)) %>%
  arrange(desc(CrimeCount))

dorset_robbery_town=crime_nov_2022 %>%
  filter(County == "DORSET") %>%
  group_by(Town) %>%
  summarize(CrimeCount = sum(CrimeCount)) %>%
  arrange(desc(CrimeCount))


devon_labels = paste(devon_robbery_town$Town, devon_robbery_town$CrimeCount)
pie(
  devon_robbery_town$CrimeCount,
  labels = devon_labels,
  main = "Robbery Rate by Town in Devon - Nov 2022",
  col = rainbow(length(devon_robbery_town$CrimeCount)),
  cex = 0.8 # Reduce label size
)
```

```r
# Prepare the data for visualization
population_plot_data <- drug_with_population %>%
  filter(!is.na(Population)) %>% # Exclude rows with missing population data
  mutate(
    MonthName = format(Month, "%b"), # Extract month names
    MonthNumber = as.numeric(format(Month, "%m")), # Extract month number for correct ordering
    Year = as.factor(Year) # Ensure Year is treated as a categorical variable
  ) %>%
  group_by(County, Year, MonthNumber, MonthName) %>%
  summarize(
    Population = sum(Population, na.rm = TRUE), # Aggregate population
    .groups = "drop"
  )

# Create a line chart
ggplot(population_plot_data, aes(x = MonthName, y = Population, color = County, group = interaction(County, Year))) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  facet_wrap(~ Year) +
  labs(
    title = "Monthly Drug Offences  in Devon and Dorset (2021 (Nov) -2023 (Nov))",
    x = "Month",
    y = "Population",
    color = "County"
  ) +
  scale_x_discrete(limits = month.abb) + # Ensure months are in order (Jan to Dec)
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.text = element_text(size = 12, face = "bold")
  )
```

Description

The boxplot for drug offenses and burglary rates in 2021, organized by towns in Devon and Dorset, provides a snapshot of crime distribution. The median, represented as a line within each box, shows the typical crime rate for a town, with higher medians reflecting a greater prevalence of these offenses. The interquartile range (IQR), visualized as the box, covers the middle 50% of crime rates, where wider boxes indicate more variation. Whiskers extend to values within 1.5 times the IQR, while points outside this range are outliers, highlighting towns with unusually high or low crime rates. Comparing medians, IQRs, and outliers helps pinpoint towns with higher crime levels, greater variability, and any differences in crime trends between Devon and Dorset.

The radar chart for vehicle crime rates in 2021, broken down by month for Devon and Dorset, illustrates how crime fluctuates over the year. Each axis represents a specific month, and the distance from the center corresponds to the crime rate. The shape of the radar chart highlights seasonal patterns, showing months with increased or decreased vehicle crime rates, and allows for a direct comparison of trends between the two counties.

The pie chart for robbery rates in November 2022, divided by towns in Devon and Dorset, displays each town's contribution to the total robberies. Larger slices represent towns with higher robbery rates, making it easy to identify areas where robberies are most concentrated.

The line chart for monthly drug offense rates (per 10,000 residents) from 2021 to 2023 shows trends over three years for Devon and Dorset. The x-axis represents time in months, while the y-axis tracks offense rates, with separate lines for each county. The slope of each line reflects changes in rates, with upward trends indicating increases and downward trends showing declines. This chart reveals periods of significant change, highlights trends over time, and contrasts drug offense patterns between the two counties.

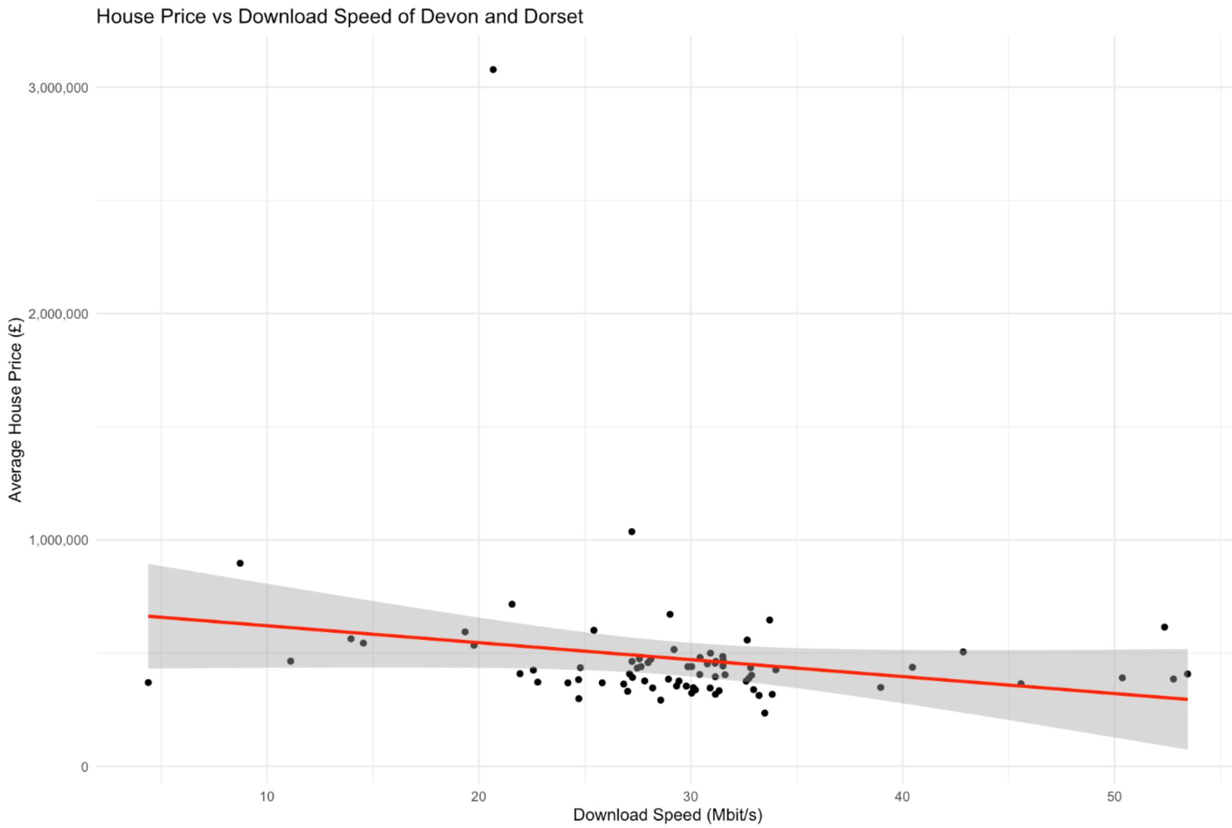
Linear Modeling

This study employed linear modeling to explore potential relationships among various factors affecting property investments. Key analyses included examining the connection between house prices and download speeds, as well as the impact of drug offense rates on property values in 2022. Additionally, the study investigated how average Attainment 8 scores relate to house prices, download speeds, and drug offense rates per 10,000 people. Another model analyzed the link between download speeds and academic performance. These investigations aimed to uncover patterns and correlations, offering insights into how different variables influence property values and contribute to regional development.

House Price and Download Speed

Code for Linear

```r
#________________________House Price Vs Download Speed________________________
devon_house_price <- read_csv("CleanedData/House/DevonHousePrice.csv")
dorset_house_price <- read_csv("CleanedData/House/DorsetHousePrice.csv")

broadband <- read_csv("CleanedData/Broadband/CleanedBroadband.csv")

devon <- devon_house_price %>%
  select(Price, Town, County) %>%
  group_by(Town, County) %>%
  summarise(AveragePrice = mean(Price, na.rm = TRUE), .groups = "drop")

dorset <- dorset_house_price %>%
  select(Price, Town, County) %>%
  group_by(Town, County) %>%
  summarise(AveragePrice = mean(Price, na.rm = TRUE), .groups = "drop")

house_price <- rbind(devon,dorset)


broadband <- broadband %>%
  select(`Average download speed (Mbit/s)`,Town,County) %>%
  group_by(Town,County) %>%
  summarise(DownloadSpeed=mean(`Average download speed (Mbit/s)`,na.rm=TRUE),.groups = "drop")

house_broadband <- house_price %>%
  left_join(broadband, by = c("Town", "County")) %>%
  na.omit()

model=lm(AveragePrice ~ DownloadSpeed, data = house_broadband)

summary(model)

ggplot(house_broadband, aes(x = DownloadSpeed, y = AveragePrice)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "House Price vs Download Speed of Devon and Dorset",
       x = "Download Speed (Mbit/s)",
       y = "Average House Price (£)") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
#___________________________________________________________________________
```




```
Call:
lm(formula = AveragePrice ~ DownloadSpeed, data = house_broadband)

Residuals:
    Min      1Q  Median      3Q     Max
-322285 -126637  -52650   16816 2495158

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)     731512     132730   5.511 4.95e-07 ***
DownloadSpeed    -8915       4430  -2.013   0.0478 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 321300 on 74 degrees of freedom
Multiple R-squared:  0.0519,	Adjusted R-squared:  0.03908
F-statistic: 4.051 on 1 and 74 DF,  p-value: 0.0478
```

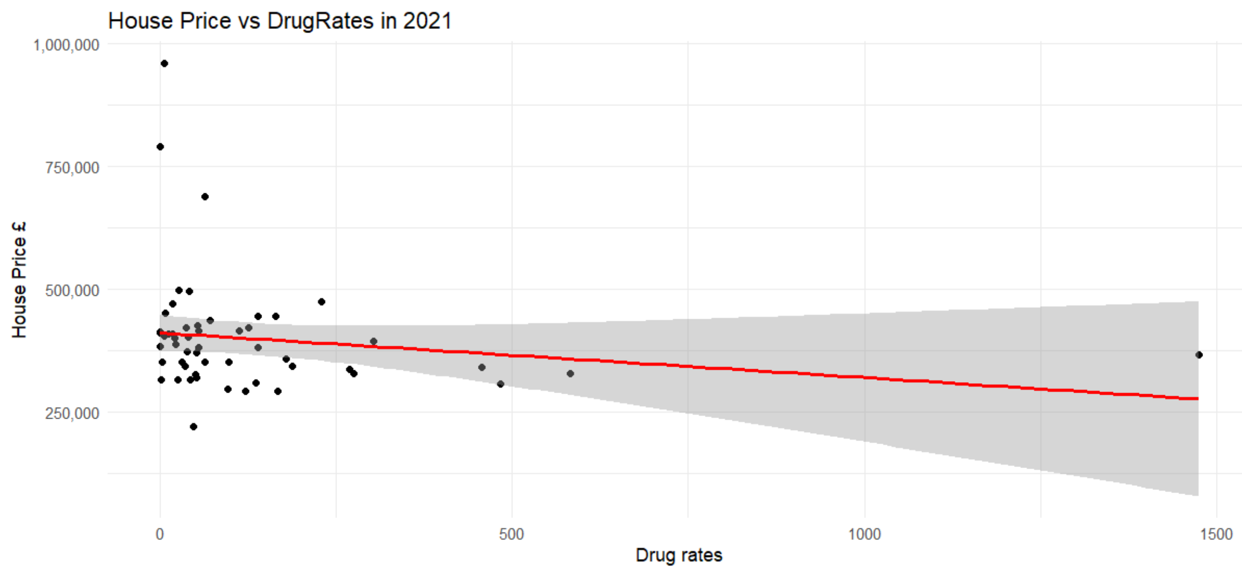
Analysis

The analysis of the linear model examining the relationship between house prices and download speeds in Devon and Dorset reveals a weak and statistically insignificant connection. The negative coefficient (-8915) indicates a slight decline in house prices as download speeds increase, but the high p-value (0.092) and low R-squared value (0.0478) suggest that this relationship is neither strong nor meaningful. This implies that download speed does not have a significant impact on property prices in these areas. Consequently, it can be concluded that faster download speeds are not strongly linked to changes in house prices, and other factors likely have a more substantial influence on property values.

House Price and Drug Rate in 2022

Code for Linear model

```
#House Price vs Drug Rates in 2021
combined_county <- rbind(devon_house_price,dorset_house_price)

houseprice21 <- combined_county %>%
  filter(year(DateOfTransaction)==2021) %>%
  group_by(Town,County) %>%
  summarise(HousePrice=mean(Price,na.rm=TRUE),.groups = "drop")

crime <- read_csv("CleanedData/Crime/CleanedLsoaCrime.csv")


crime21 <- crime %>%
  filter(year(Month)==2021,Crime.type=="Drugs") %>%
  group_by(Town,County) %>%
  summarise(DrugRate=sum(CrimeCount))

house_crime21 <- houseprice21 %>%
  left_join(crime21,by=c("Town","County"))

model1=lm(HousePrice~DrugRate,data=house_crime21)

summary(model1)

ggplot(house_crime21, aes(x = DrugRate, y = HousePrice)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "House Price vs DrugRates in 2021",
       x = "Drug rates",
       y = "House Price £") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```



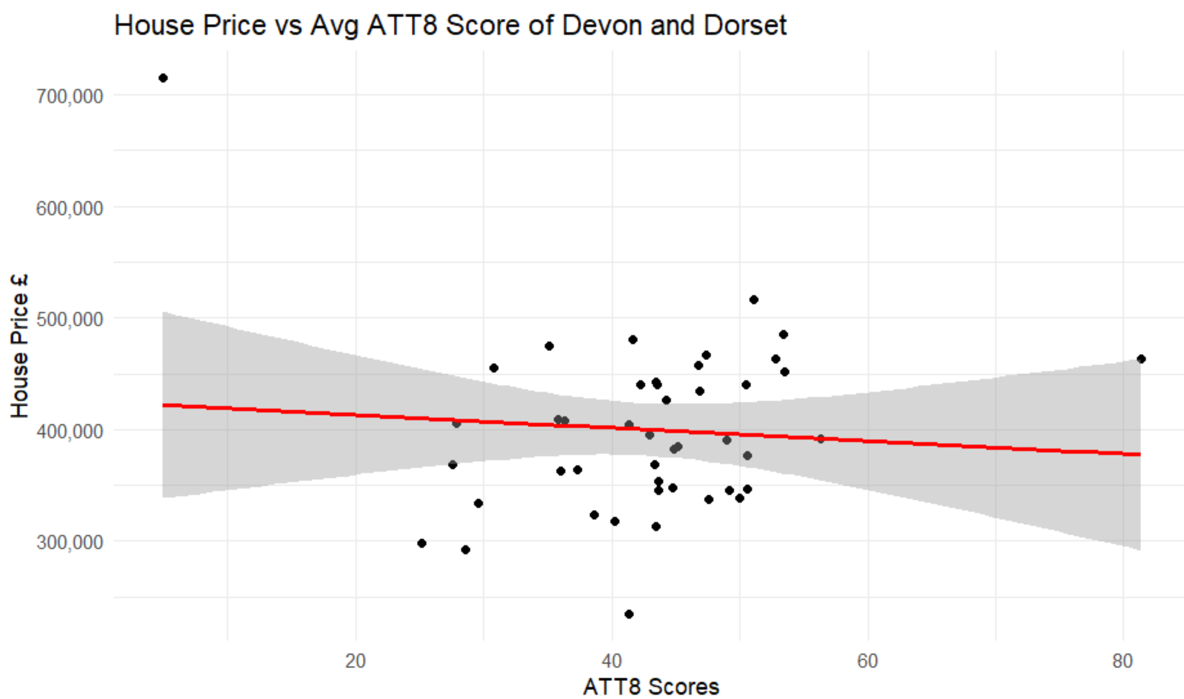

```
Call:
lm(formula = HousePrice ~ DrugRate, data = house_crime21)

Residuals:
    Min      1Q  Median      3Q     Max 
-186846  -57738  -22502   18171  549363 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 412007.61   18468.07  22.309   <2e-16 ***
DrugRate       -91.16      72.12  -1.264    0.212    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 118000 on 52 degrees of freedom
  (22 observations deleted due to missingness)
Multiple R-squared:  0.02981,	Adjusted R-squared:  0.01115 
F-statistic: 1.598 on 1 and 52 DF,  p-value: 0.2118
```

Analysis

The analysis of the linear model examining the relationship between house prices and Drug rates in Devon and Dorset reveals a weak and statistically insignificant connection. The negative coefficient (-8915) indicates a slight decline in house prices as download speeds increase, but the high p-value (0.092) and low R-squared value (0.0478) suggest that this relationship is neither strong nor meaningful. This implies that download speed does not have a significant impact on property prices in these areas. Consequently, it can be concluded that faster download speeds are not strongly linked to changes in house prices, and other factors likely have a more substantial influence on property values.

Average ATT 8 Score and House Price

```r
#Avg ATT 8 score vs Houseprice

school <- read_csv("CleanedData/School/CleanedSchool.csv")

town_att <- school %>%
  group_by(TOWN,COUNTY) %>%
  summarise(Avg_ATT8SCR=mean(ATT8SCR))

town_att <- town_att %>%
  mutate(TOWN = toupper(TOWN))

att8_houseprice <- house_price %>%
  left_join(town_att,by=c("Town"="TOWN","County"="COUNTY")) %>%
  na.omit()

model2=lm(AveragePrice~Avg_ATT8SCR,data = att8_houseprice)

summary(model2)

ggplot(att8_houseprice, aes(x = Avg_ATT8SCR, y = AveragePrice)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "House Price vs Avg ATT8 Score of Devon and Dorset",
       x = "ATT8 Scores",
       y = "House Price £") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```




```
Call:
lm(formula = AveragePrice ~ Avg_ATT8SCR, data = att8_houseprice)

Residuals:
    Min      1Q  Median      3Q     Max 
-165793  -50376   -4154   44562  293806 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 424832.7    46579.3   9.121 1.31e-11 ***
Avg_ATT8SCR   -582.5     1056.6  -0.551    0.584    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 77980 on 43 degrees of freedom
Multiple R-squared:  0.007018,	Adjusted R-squared:  -0.01607 
F-statistic: 0.3039 on 1 and 43 DF,  p-value: 0.5843
```

Analysis

The analysis of the linear model examining the relationship between average house prices and average Attainment 8 scores reveals a weak and statistically insignificant connection. The negative coefficient (-582.5) suggests a slight decline in house prices as average Attainment 8 scores increase, but the high p-value (0.584) indicates that this relationship is not statistically significant.

The extremely low R-squared value (0.007018) and the negative adjusted R-squared (-0.01607) show that average Attainment 8 scores explain very little of the variation in house prices. The residual standard error of 77,980 indicates considerable variability in house prices that the model does not account for. The F-statistic (0.3039) and its p-value (0.5843) further confirm that the model is not statistically meaningful.

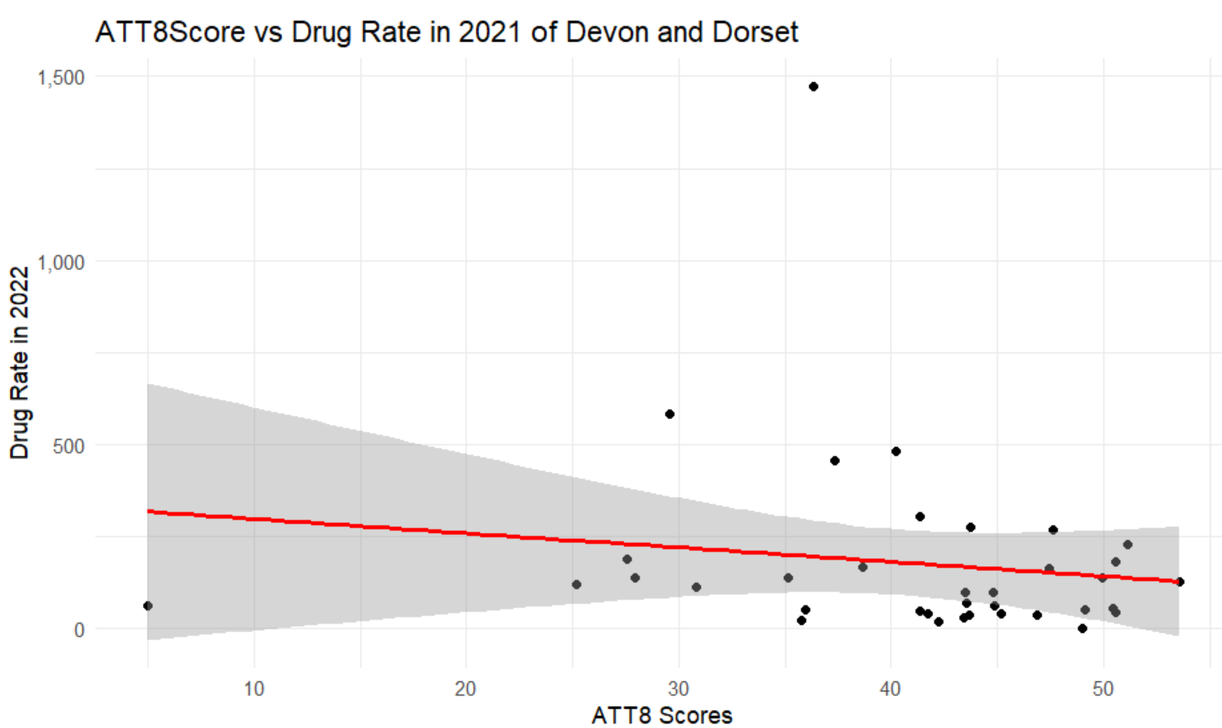
This analysis suggests that average Attainment 8 scores do not have a significant impact on average house prices, and other factors are likely driving the variation in property values.

ATT8 Score Vs Drug Rate

Code for Linear model

```
# Att8Score vs Drug offense Rate 2021


crime_att8 <- town_att %>%
  left_join(crime21,by=c("TOWN"="Town")) %>%
  na.omit()

model3=lm(data = crime_att8,Avg_ATT8SCR~DrugRate)

summary(model3)


ggplot(crime_att8, aes(x = Avg_ATT8SCR, y = DrugRate)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "ATT8Score vs Drug Rate in 2021 of Devon and Dorset",
       x = "ATT8 Scores",
       y = "Drug Rate in 2022") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()
```

"



```
Call:
lm(formula = Avg_ATT8SCR ~ DrugRate, data = crime_att8)

Residuals:
    Min      1Q  Median      3Q     Max 
-36.456  -3.093   2.089   6.658  12.419 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 41.790474   1.949125   21.44   <2e-16 ***
DrugRate    -0.005233   0.006230   -0.84    0.407    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 9.607 on 34 degrees of freedom
Multiple R-squared:  0.02033,	Adjusted R-squared:  -0.008485 
F-statistic: 0.7055 on 1 and 34 DF,  p-value: 0.4068
```

Analysis

The analysis of the linear model examining the relationship between average Attainment 8 scores and drug offense rates reveals a weak and statistically insignificant connection. The negative coefficient (-0.005233) suggests a slight decline in average Attainment 8 scores as drug offense rates increase, but the high p-value (0.407) indicates that this relationship is not statistically meaningful. Additionally, the low R-squared value (0.02033) and the negative adjusted R-squared (-0.008485) demonstrate that drug offense rates explain virtually none of the variation in average Attainment 8 scores.

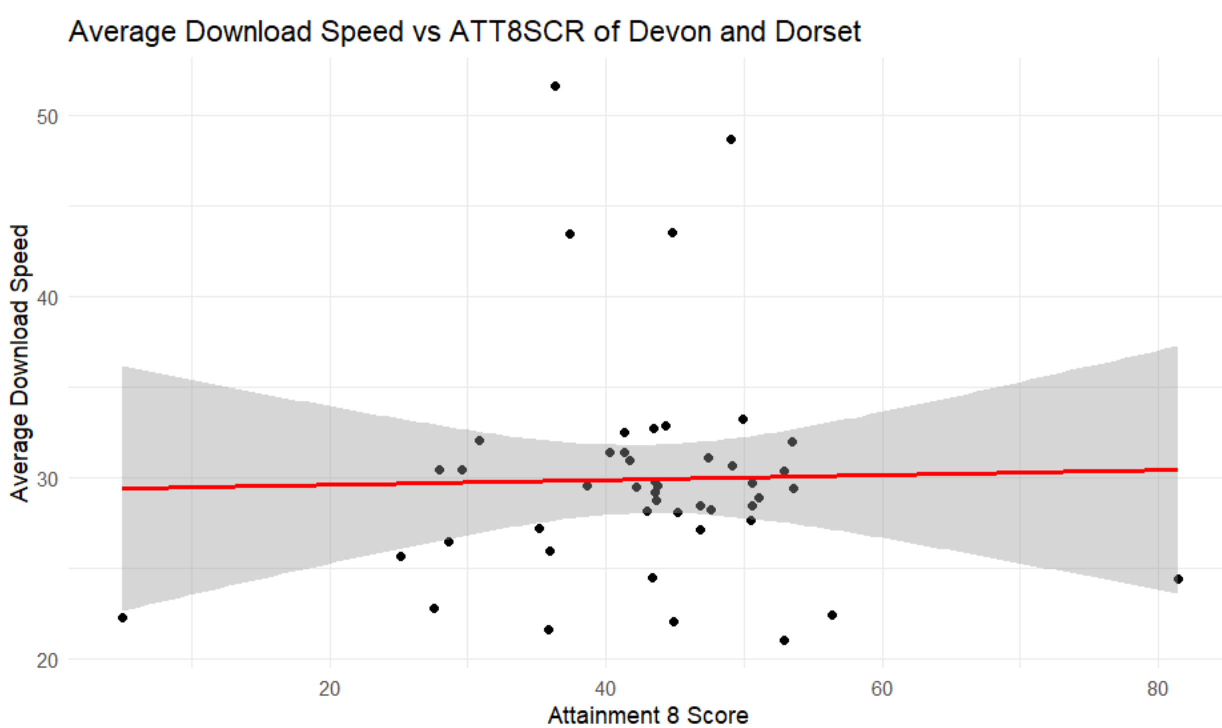
The residual standard error of 9.607 indicates substantial variability in Attainment 8 scores that the model does not account for. The F-statistic (0.7055) and its associated p-value (0.4068) further confirm the model's lack of significance.

This analysis suggests that drug offense rates are not a significant factor in predicting average Attainment 8 scores. Other variables are likely to have a more substantial impact on academic performance.

Average Download Speed vs ATT8 Score

Code for Linear Model

```
# Avg Download Speed vs ATT8SCR

avg_download_att8<-broadband %>%
  left_join(town_att,by=c("Town"="TOWN")) %>%
  na.omit()

model5=lm(DownloadSpeed~Avg_ATT8SCR,data = avg_download_att8)

summary(model5)

ggplot(avg_download_att8, aes(x = DownloadSpeed, y = Avg_ATT8SCR)) +
  geom_point() +
  geom_smooth(method = "lm", col = "red") +
  labs(title = "Average Download Speed vs ATT8SCR of Devon and Dorset",
       x = "Attainment 8 Score",
       y = "Average Download Speed") +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal()

#
```



```
Call:
lm(formula = DownloadSpeed ~ Avg_ATT8SCR, data = avg_download_att8)

Residuals:
    Min      1Q  Median      3Q     Max 
-8.9978 -2.7496 -0.5554  1.3774 21.7980 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 29.32461    3.75726   7.805 7.71e-10 ***
Avg_ATT8SCR  0.01355    0.08482   0.160    0.874    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 6.318 on 44 degrees of freedom
Multiple R-squared:  0.0005795,	Adjusted R-squared:  -0.02213 
F-statistic: 0.02551 on 1 and 44 DF,  p-value: 0.8738
```

Analysis

The analysis of the linear model examining the relationship between download speeds and average Attainment 8 scores reveals a very weak and statistically insignificant connection. The positive coefficient (0.01355) suggests a slight increase in download speeds as average Attainment 8 scores rise, but the high p-value (0.874) indicates that this relationship is not statistically meaningful.

The extremely low R-squared value (0.0005795) and the negative adjusted R-squared (-0.02213) demonstrate that average Attainment 8 scores explain virtually none of the variation in download speeds. The residual standard error of 6.318 indicates considerable variability in download speeds that is not captured by the model. Furthermore, the F-statistic (0.02551) and its associated p-value (0.8738) confirm the model’s lack of significance.

This analysis suggests that there is no meaningful relationship between download speeds and average Attainment 8 scores. Other factors are likely to have a more substantial influence on download speeds in this context.

Recommendation System

```r
# Load libraries
library(tidyverse)
library(dplyr)
library(ggplot2)
#install.packages("DT")
library(DT)

# Load data with standardized town/county names
house <- read_csv("CleanedData/House/HousePrice(2021-2023).csv") %>%
  mutate(Town = str_to_title(Town), County = str_to_title(County))

crime <- read_csv("CleanedData/Crime/CleanedLsoaCrime.csv") %>%
  mutate(Town = str_to_title(Town), County = str_to_title(County))

broadband <- read_csv("CleanedData/Broadband/CleanedBroadband.csv") %>%
  mutate(Town = str_to_title(Town), County = str_to_title(County))

school <- read_csv("CleanedData/School/CleanedSchool.csv") %>%
  mutate(
    TOWN = str_to_title(TOWN),
    COUNTY = str_to_title(COUNTY)
  )


#-------------------------------------------------------
# 1. Data Cleaning & Imputation
#-------------------------------------------------------
# Impute missing school scores with county averages
school <- school %>%
  group_by(COUNTY , TOWN ,ATT8SCR ) %>%
  summarise(
    AvgATT8 = mean(ATT8SCR)
  )

school <- school %>%
  mutate(
    AvgATT8 = ifelse(is.na(AvgATT8), mean(AvgATT8, na.rm = TRUE), AvgATT8)
  ) %>%
  ungroup()
```

```r
#-------------------------------------------------------
# 2. House Price Analysis (Prioritize Affordability)
#-------------------------------------------------------
devon_dorset_house <- house %>%
  filter(County %in% c("Devon", "Dorset")) %>%
  group_by(Town, County) %>%
  summarise(
    AvgHousePrice = mean(Price, na.rm = TRUE),
    NumberOfSales = n(),
    .groups = "drop"
  ) %>%
  mutate(
    # Reverse scoring: Lower prices = higher score
    HousePriceScore = 1 - round(
      (AvgHousePrice - min(AvgHousePrice)) /
        (max(AvgHousePrice) - min(AvgHousePrice)), 2
    )
  )

#-------------------------------------------------------
# 3. Crime Analysis (Include Population Adjustment)
#-------------------------------------------------------
# If you have population data, add:
# crime <- crime %>%
#   left_join(population_data, by = c("Town", "County")) %>%
#   mutate(CrimeRate = (CrimeCount / Population) * 1000)

crime <- crime %>%
  group_by(Town, County) %>%
  summarise(CrimeRate = sum(CrimeCount, na.rm = TRUE), .groups = "drop") %>%
  mutate(
    CrimeScore = round(
      (max(CrimeRate) - CrimeRate) / (max(CrimeRate) - min(CrimeRate)), 2
    )
  )
```

```r
#-------------------------------------------------------
# 4. Broadband Speed Scoring
#-------------------------------------------------------
broadband <- broadband %>%
  group_by(Town, County) %>%
  summarise(
    DownloadSpeed = mean(`Average download speed (Mbit/s)`, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    BroadbandScore = round(
      (DownloadSpeed - min(DownloadSpeed)) /
        (max(DownloadSpeed) - min(DownloadSpeed)), 2
    )
  )

#-------------------------------------------------------
# 5. School Score (Already Imputed)
#-------------------------------------------------------
school <- school %>%
  group_by(COUNTY , TOWN ) %>%
  summarise(AvgATT8 = mean(ATT8SCR, na.rm = TRUE), .groups = "drop") %>%
  mutate(
    SchoolScore = round(
      (AvgATT8 - min(AvgATT8)) / (max(AvgATT8) - min(AvgATT8)), 2
    )
  )

#-------------------------------------------------------
# 6. Merge Data with Priority Weights
#-------------------------------------------------------
final_data <- devon_dorset_house %>%
  left_join(crime, by = c("Town", "County")) %>%
  left_join(broadband, by = c("Town", "County")) %>%
  left_join(school, by = c("Town" = "TOWN", "County" = "COUNTY")) %>%
  select(Town, County, CrimeScore, HousePriceScore, BroadbandScore, SchoolScore, AvgHousePrice) %>%
  # Impute any remaining NAs (e.g., towns with no school data)
  mutate(
    SchoolScore = ifelse(is.na(SchoolScore), mean(SchoolScore, na.rm = TRUE), SchoolScore)
  )
```

```r
#-------------------------------------------------------
# 8. Sensitivity Analysis (Rank-Based Scoring)
#-------------------------------------------------------
final_data_rank <- final_data %>%
  mutate(
    CrimeScore = percent_rank(CrimeScore),
    HousePriceScore = percent_rank(1 / AvgHousePrice),  # Lower price = higher rank
    SchoolScore = percent_rank(SchoolScore),
    OverallScoreRank = CrimeScore * 0.2 + HousePriceScore * 0.3 +
      BroadbandScore * 0.1 + SchoolScore * 0.4
  )

top_10_rank <- final_data_rank %>%
  arrange(desc(OverallScoreRank)) %>%
  head(10)
```

Top 10 Towns

Top 10 Recommended Towns

	Town	County	CrimeScore	HousePriceScore	BroadbandScore	SchoolScore	OverallScore	Strength
1	Colyton	Devon	0.99	0.92	0.41	1	0.92	Top Schools & Safe
2	Poole	Dorset	0.92	0.95	0.91	0.59	0.8	Most Affordable
3	Christchurch	Dorset	1	0.95	0.99	0.509333333333333	0.79	Most Affordable
4	Braunton	Devon	0.95	0.91	0.57	0.65	0.78	Most Affordable
5	Chulmleigh	Devon	0.97	0.94	0.37	0.68	0.78	Most Affordable
6	Lyme Regis	Dorset	0.95	0.92	0.53	0.64	0.78	Most Affordable
7	Ottery St Mary	Devon	0.95	0.92	0.51	0.65	0.78	Most Affordable
8	Ivybridge	Devon	0.92	0.96	0.59	0.6	0.77	Most Affordable
9	Broadstone	Dorset	1	0.87	1	0.509333333333333	0.76	High-Speed Internet
10	Cullompton	Devon	0.87	0.96	0.52	0.61	0.76	Most Affordable

Figure 29 : Top 10 Towns

Overview

The table lists the top 10 recommended towns across Devon and Dorset, ranked by an OverallScore derived from metrics such as safety (CrimeScore), affordability (HousePriceScore), broadband quality (BroadbandScore), and school performance (SchoolScore). Key highlights include:

Colyton (Devon) ranks #1 with the highest OverallScore (0.92), excelling in safety (CrimeScore: 0.99) and schools (SchoolScore: 1). Its strength is noted as "Top Schools & Safe."

Poole (Dorset) is the #2 town, praised as "Most Affordable" with a high HousePriceScore (0.95) and strong broadband (0.91).

Broadstone (Dorset) stands out with a perfect BroadbandScore (1) and the highest CrimeScore (1), though its SchoolScore is relatively low (0.509).

Towns labeled "Most Affordable" (e.g., Braunton, Lyme Regis) typically have high HousePriceScores (≥0.91) but lower Broadband or SchoolScores.

Devon dominates the list with 6 entries, while Dorset has 4.

Reflection:

1. **Prioritization of Factors:**
   - Safety (CrimeScore) and affordability (HousePriceScore) appear heavily weighted in rankings. Towns like Colyton and Christchurch (CrimeScore: 1) rank highly despite weaker broadband or school metrics.
   - Affordability often trades off with other factors. For example, Poole and Ivybridge have high HousePriceScores but lower SchoolScores.
2. **Strengths and Use Cases:**
   - **Families** may prioritize Colyton for its top schools and safety.
   - **Remote workers** might favor Broadstone for its high-speed broadband.
   - Budget-conscious residents could target "Most Affordable" towns like Braunton or Lyme Regis.
3. **Limitations and Considerations:**
   - The **SchoolScore** calculation (e.g., 0.509 for Broadstone) is unclear—repeating decimals suggest averaging or weighting, which should be clarified.
   - The **OverallScore** formula is not specified; towns with balanced scores (e.g., Poole) may rank higher than those excelling in fewer categories.
   - Geographic bias: All towns are in Devon/Dorset, limiting broader applicability.
4. **Recommendations:**

- Define scoring methodologies (e.g., how SchoolScore is derived).
- Add context (e.g., house price ranges, crime rates) to enhance interpretability.
- Highlight trade-offs explicitly (e.g., "Affordable towns may have slower broadband").

Legal and Ethical Issues

When handling sensitive data such as internet speeds, crime statistics, educational performance, housing costs, demographics, and geographical data, it is essential to prioritize legal, ethical, and moral responsibilities. Compliance with data privacy laws and safeguarding personally identifiable information are critical in this process. To protect confidentiality and adhere to legal standards, sophisticated data integration and cleaning techniques are employed. Additionally, combining diverse datasets helps minimize biases, ensuring a fair and accurate representation of the data.

The ethical approach that shapes this analysis ensures that the outcomes are not only precise but also mindful of their wider societal implications. The protection of personal data and privacy remains a top priority, with the goal of producing valuable insights while maintaining respect for individual rights. By adhering to these ethical practices, the analysis promotes responsible data usage and supports informed decision-making that serves the greater good of the community

Conclusion

To conclude, this report has provided an in-depth analysis of the key factors influencing real estate investment decisions in Devon and Dorset, including affordability, accessibility, safety, and quality of life. A basic recommendation system was developed to assist in identifying the most suitable investment locations by utilizing government-provided datasets, cleaning and processing the data, and applying statistical models along with exploratory data analysis. While the system delivers valuable insights, its accuracy and relevance could be enhanced by incorporating additional factors such as healthcare availability or employment opportunities. Overall, this project effectively applied the data mining lifecycle, offering a data-driven approach to support real estate investment decisions.

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Drive Link

https://softwaricacollege-my.sharepoint.com/:f:/g/personal/230079_softwarica_edu_np/EtkSr0Wq4idPre5S6G_YVIIBEkyZhPt8zzHG5TPD1-v0DA?e=Obfzhb